Name:____________

GCSE Chemistry Unit 1 Foundation Tier

Once you have completed a paper and have marked it, fill out the table below, using the grade boundaries provided to add your grade. Then list which questions you did well and which questions you struggled with. For the questions your struggled with, review why and at the reasons. The most common reasons are

- Not understanding basic concepts (e.g. simple definitions)
- Calculation errors (forgetting to square numbers, etc.)
- Not reading questions carefully
- Not answering the full question (describe vs. explain)
- Laboratory/practical setup mistakes

For these questions it is also useful add notes to the question to help you next time and take a photo of these questions. Use these then to guide further revision.

Year	Score /80	Grade	Good Questions	Bad Questions	Reasons
2017					
2018					
2019					
2022					
2023					

Grade boundaries - Raw Score

Year	G	F	E	D	C	Max UMS
2017	18	23	28	34	40	46
2018	13	19	25	31	37	43
2019	14	19	24	29	35	41
2022	10	15	20	25	30	36
2023	13	21	29	37	46	55

Surname	Centre Number	Candidate Number
Other Names		0

GCSE – NEW

3410U10-1

S17-3410U10-1

CHEMISTRY – Unit 1:
Chemical Substances, Reactions and Essential Resources

FOUNDATION TIER

FRIDAY, 16 JUNE 2017 – MORNING

1 hour 45 minutes

For Examiner's use only		
Question	**Maximum Mark**	**Mark Awarded**
1.	**8**	
2.	**13**	
3.	**7**	
4.	**8**	
5.	**8**	
6.	**6**	
7.	**10**	
8.	**9**	
9.	**11**	
Total	**80**	

3410U101
01

ADDITIONAL MATERIALS

In addition to this paper you may require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

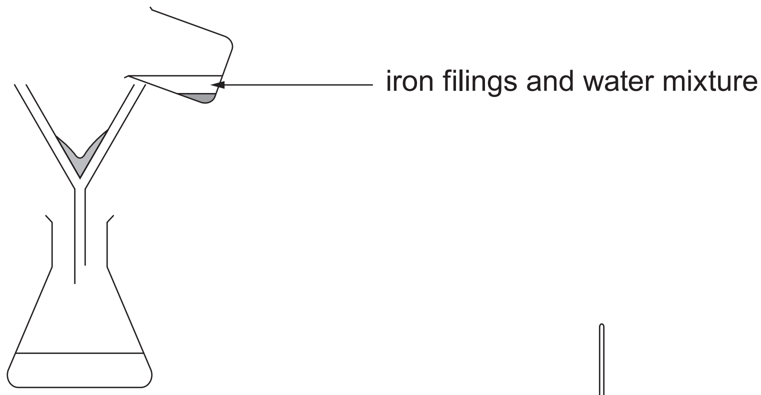
Question **6** is a quality of extended response (QER) question where your writing skills will be assessed.

The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.

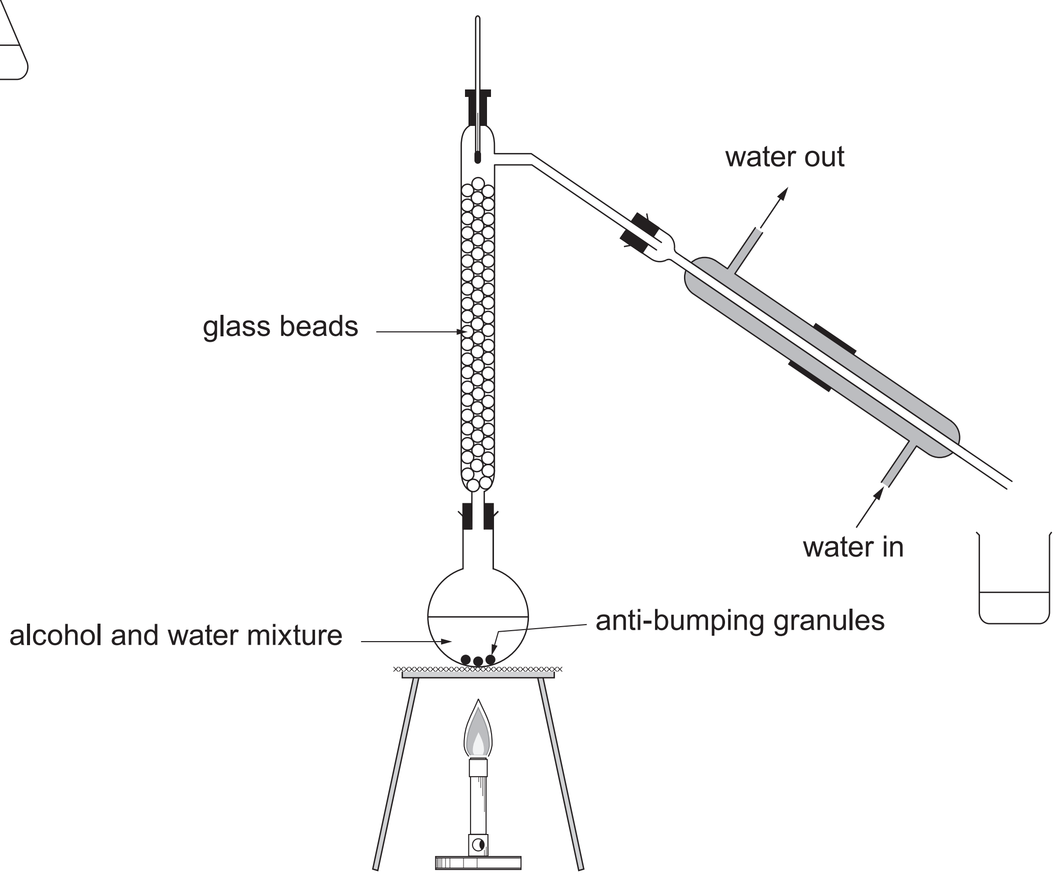
Answer **all** *questions*.

1. *(a)* The diagrams show three methods used to separate mixtures.

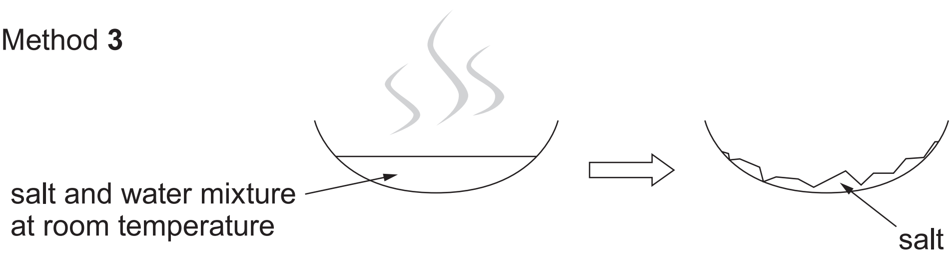
Method **1**



Method **2**



Method **3**



Examiner only

(i) Underline the property of iron filings which allows method **1** to be used to separate the mixture. [1]

magnetic **insoluble in water** **silvery grey colour**

high melting point **more dense than water**

(ii) Tick (✓) the box that shows the reason why method **2** can be used to separate alcohol and water. [1]

alcohol and water both boil when heated ☐

alcohol and water have different boiling points ☐

alcohol and water have the same boiling point ☐

alcohol and water are both liquids ☐

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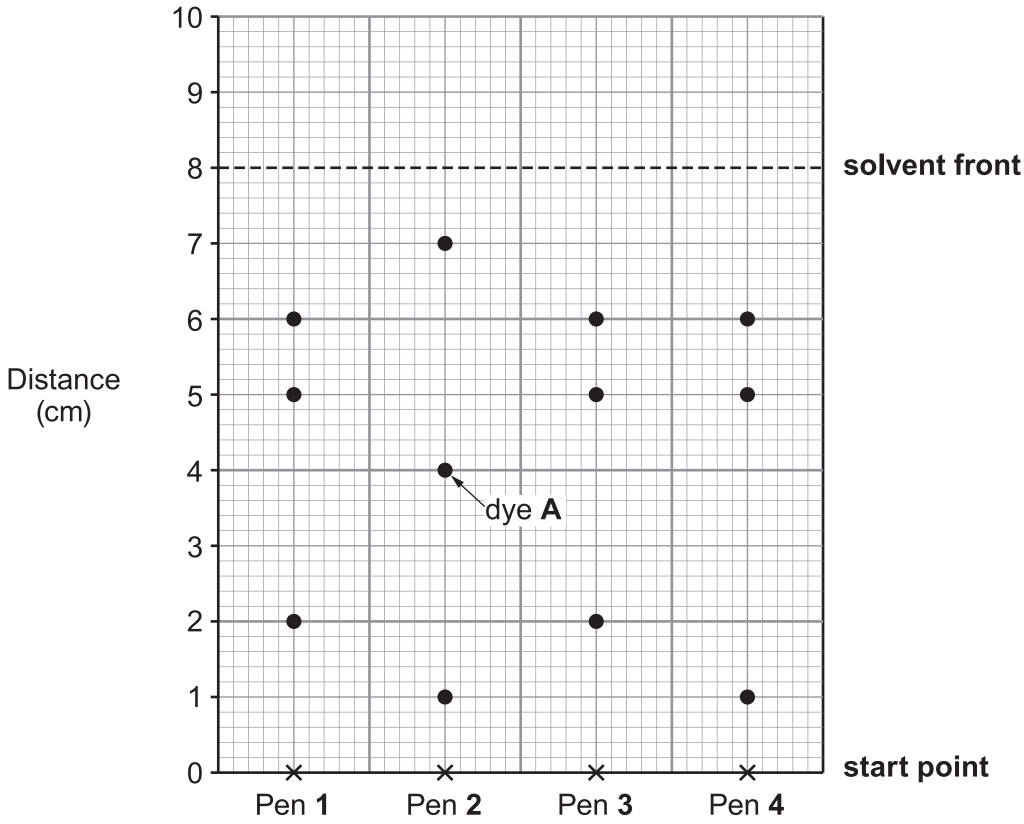
(iii) Name the process of removing water in method **3**. [1]

....

Examiner only

(b) A group of students were asked to investigate the dyes in four different water soluble blue pens, **1**, **2**, **3** and **4**. The results are shown below.



(i) Which **two** pens contain the same three dyes? [1]

Pens .... and ....

(ii) Which pen contains the **most** soluble dye? [1]

Pen ....

(iii) The R_f value of a substance can be used to identify that substance.

The R_f value is given by the formula:

$$R_f = \frac{\text{distance moved by the substance}}{\text{distance moved by the solvent front}}$$

Calculate the R_f value for dye **A**. [2]

R_f = ...

Examiner only

(iv) The box contains methods for separating mixtures.

distillation	**chromatography**	**filtration**

Complete the sentence using a method from the box.

The method used to separate the coloured dyes in the blue pens is called

……………………………………………………………………………. [1]

8

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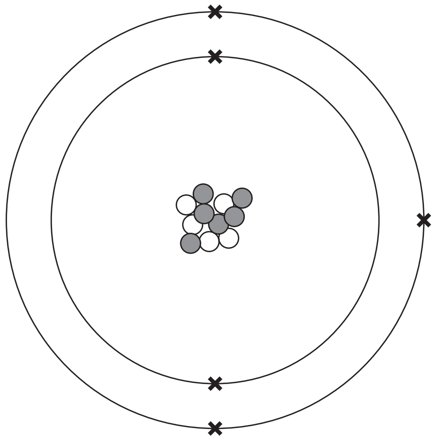
2. *(a)* Atoms contain particles called electrons, protons and neutrons. The diagram shows an atom of boron.



Complete the sentences. [4]

The nucleus of a boron atom contains five .... and

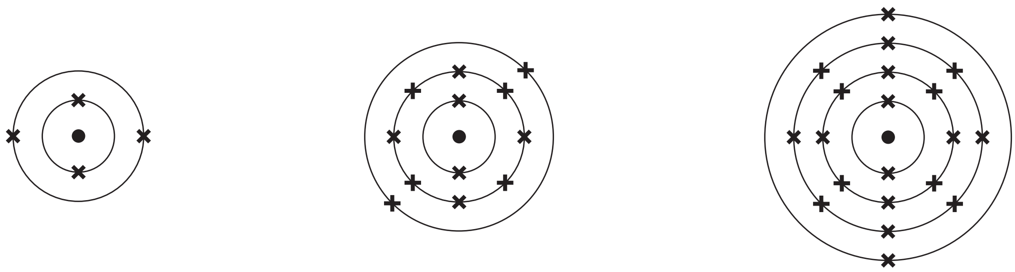
six ....

There are five .... in the shells around the nucleus.

The particles in the atom which have a positive charge are called

....

(b) The diagrams show three different atoms labelled **A**, **B** and **C**.
These letters are **not** chemical symbols.



A **B** **C**

Complete the sentences.

(i) Atoms **A**, **B** and **C** can be found in Group .... [1]

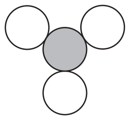
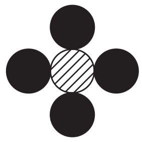
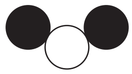
(ii) The atomic number of atom **A** is .... [1]

(iii) The electronic structure of atom **C** is (2, ...). [1]

(iv) Atom **B** can be found in Period .... [1]

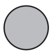
Examiner only

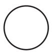
(c) The following table shows three substances, their formulae and diagrams that can be used to represent them.

Substance	Formula	Diagram
nitrogen trioxide	NO_3	
methane	CH_4	
water	H_2O	

(i) Use the information in the table to work out the key being used to represent the different elements in the diagrams. [2]

 is ....

 is ....

 is ....

 is ....

(ii) **Using the same key** draw a diagram to represent a molecule of carbon dioxide, CO_2. [1]

(d) The chemical formula of phosphoric acid is H_3PO_4.

(i) State how many phosphorus atoms are present in the formula H_3PO_4. .... [1]

(ii) Give the **total** number of atoms shown in the formula H_3PO_4. .... [1]

13

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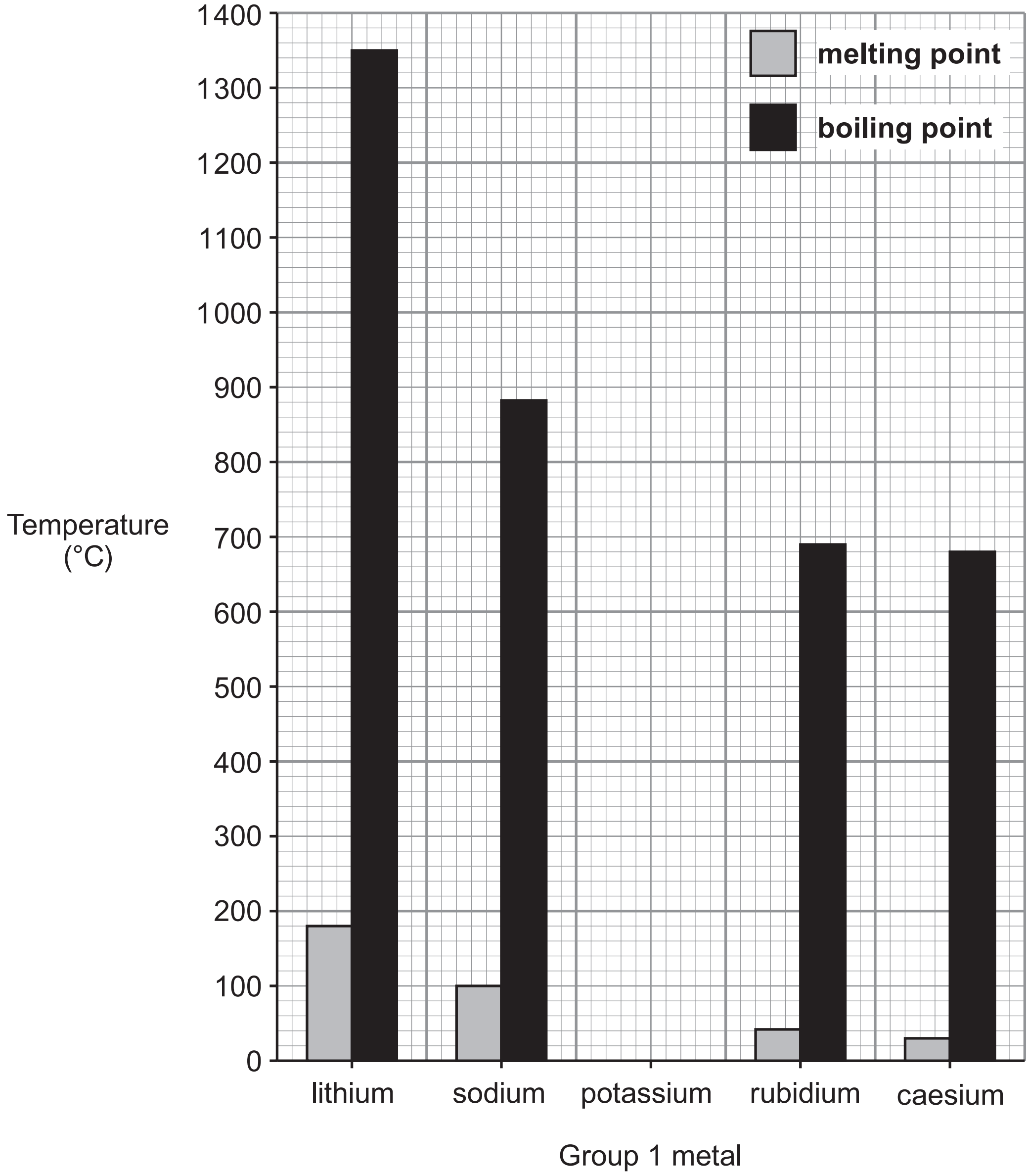
3. *(a)* The chart below shows the melting points and boiling points of some Group 1 metals. The data for potassium is missing.



Tick (✓) the box that shows the melting point and boiling point values for potassium which fit the trends shown in the chart. [1]

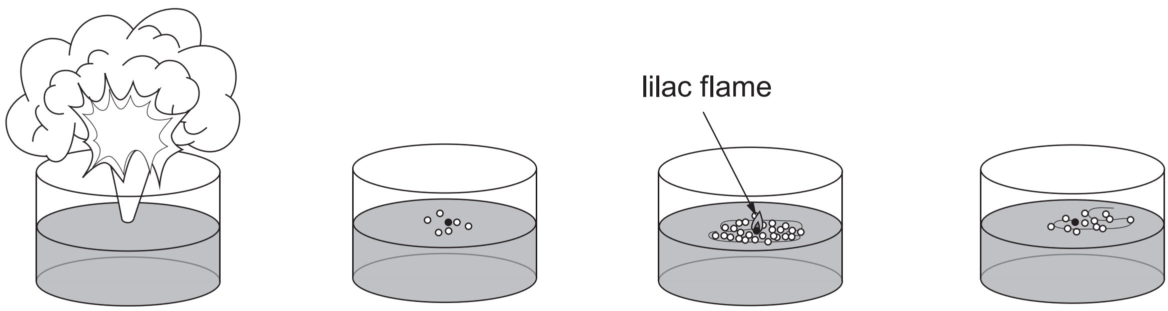
melting point = 40 °C
boiling point = 600 °C ☐

melting point = 60 °C
boiling point = 780 °C ☐

melting point = 20 °C
boiling point = 600 °C ☐

Examiner only

(b) The diagrams show lithium, sodium, potassium and rubidium, **but not necessarily in that order**, reacting separately with cold water.



A	B	C	D
Violent reaction with sparks spitting out of the trough	Gentle fizzing	Metal ball melts, moves around on the surface and burns	Metal ball fizzes and moves around on the surface

(i) Use the information in the diagrams to give the **letter** which represents: [2]

lithium ....

sodium ....

potassium ....

rubidium ....

(ii) Caesium lies below rubidium in Group 1. Suggest how its reaction with water would be different from those above. [1]

....

(iii) Describe **one** safety precaution taken when adding a Group 1 metal to water. [1]

....

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Examiner only

(c) Group 1 metals react with oxygen to form metal oxides.

Sodium oxide contains the ions Na^+ and O^{2-}.

Underline the correct formula of sodium oxide. [1]

2NaO **NaO_2** **Na_2O** **Na^2O**

(d) Group 1 metals react with chlorine, Cl_2, to form metal chlorides.

Give the **letter** for the balanced symbol equation for the reaction between sodium and chlorine. [1]

A $Na + Cl \longrightarrow NaCl$

B $2Na + 2Cl \longrightarrow 2NaCl$

C $2Na + Cl_2 \longrightarrow 2NaCl$

D $Na + Cl_2 \longrightarrow NaCl_2$

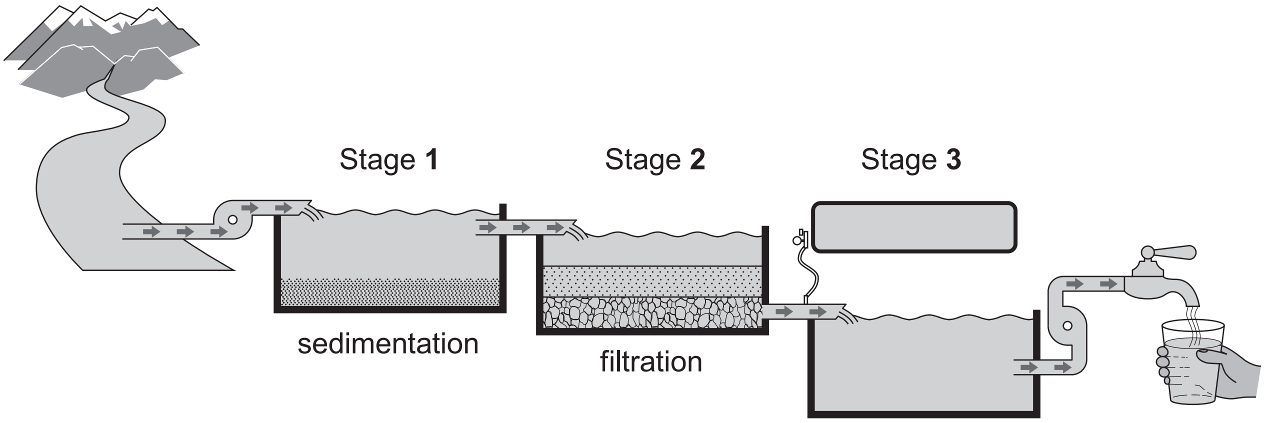
Letter ....

7

Examiner only

4. *(a)* Most water in Wales is sourced from surface water and ground water. Ground water is the water beneath the surface of the Earth, consisting mainly of surface water that has filtered through the rocks. It is the source of water found in springs and wells. Human activity causes surface water to be more contaminated than ground water.

The diagram below shows the three main stages in the treatment of surface water before it enters the mains water system on its way to the taps in your home.



(i) Tick (✓) the box that describes what occurs in stage **1**. [1]

large insoluble particles sink to the bottom	☐
soluble particles are removed	☐
small fine insoluble particles are removed	☐
fluoride is added to the water	☐

(ii) Stage **3** makes water safe to drink. Name the substance added to water in stage **3**. State why this makes the water safe to drink. [2]

Substance ....

....

(iii) Describe **one major** cause of surface water pollution. [2]

....

....

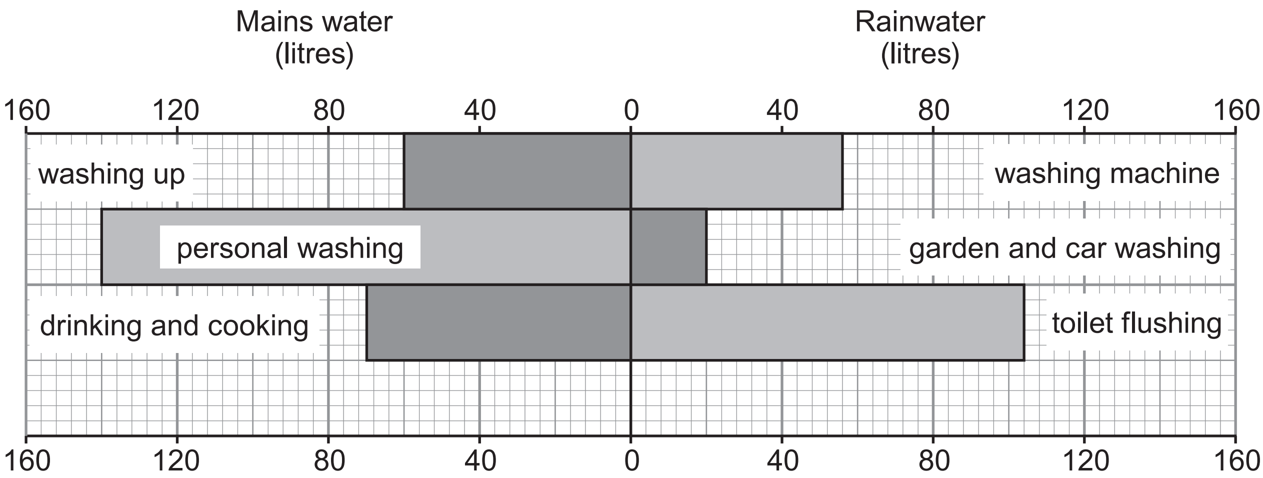
....

....

Examiner only

(b) A family uses a water butt to collect rainwater in order to reduce their use of mains water.

They use a total of 450 litres of water on a given day. The chart shows the volumes of mains water and collected rainwater used for various purposes on that day.



(i) What percentage of the total water used on that day is rainwater? [2]

percentage rainwater = ... %

(ii) Suggest why it is safe to use rainwater for the uses shown on the right hand side of the chart. [1]

....

....

8

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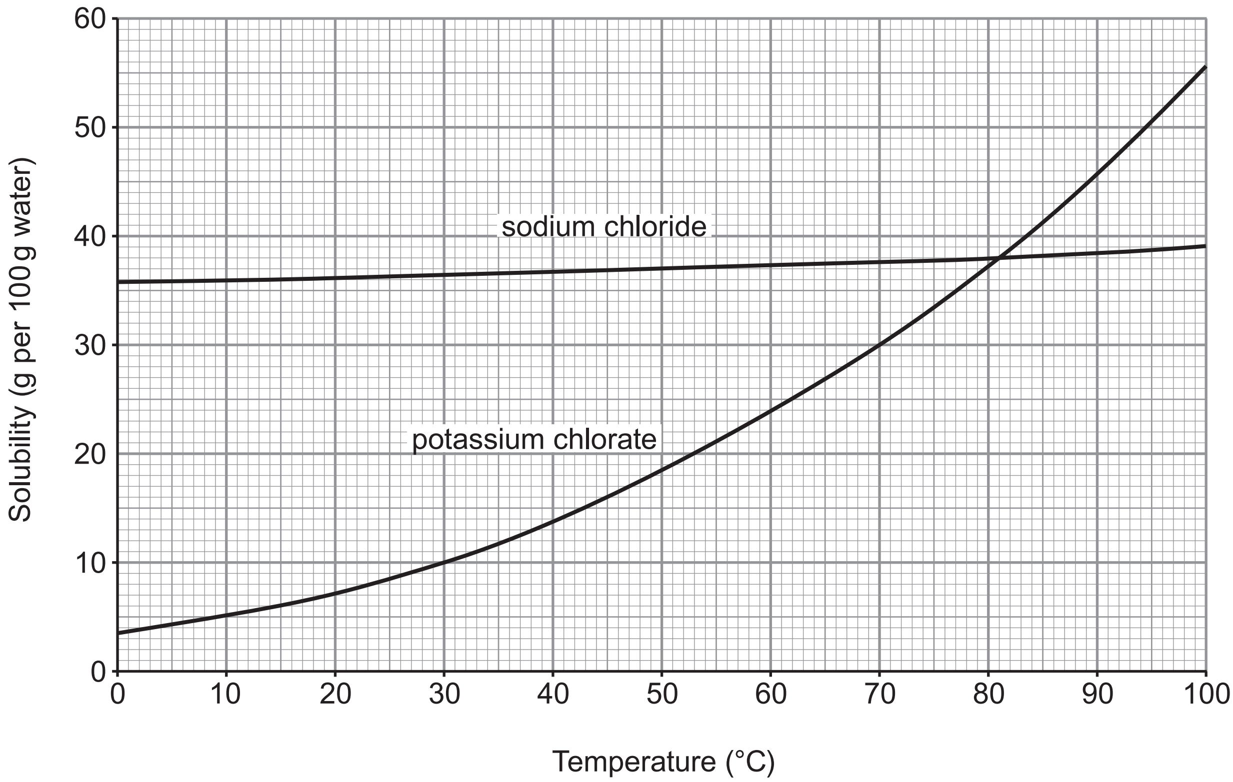
Examiner only

5. *(a)* The graphs below show the solubilities of sodium chloride and potassium chlorate in water at different temperatures.



(i) State the temperature at which the two compounds have the same solubility. [1]

.... °C

(ii) Calculate the mass of solid potassium chlorate that forms when a saturated solution in 100 g of water at 70 °C cools to 30 °C. [2]

mass = ... g

Examiner only

(b) (i) Calculate the relative formula mass (M_r) of potassium chlorate, $KClO_3$. [2]

relative formula mass = ...

(ii) Calculate the percentage by mass of potassium in potassium chlorate. [2]

percentage by mass = ... %

(c) Potassium chlorate forms potassium chloride and oxygen on heating.

Balance the symbol equation that represents the reaction taking place. [1]

$$2KClO_3 \longrightarrow \boxed{}\, KCl + \boxed{}\, O_2$$

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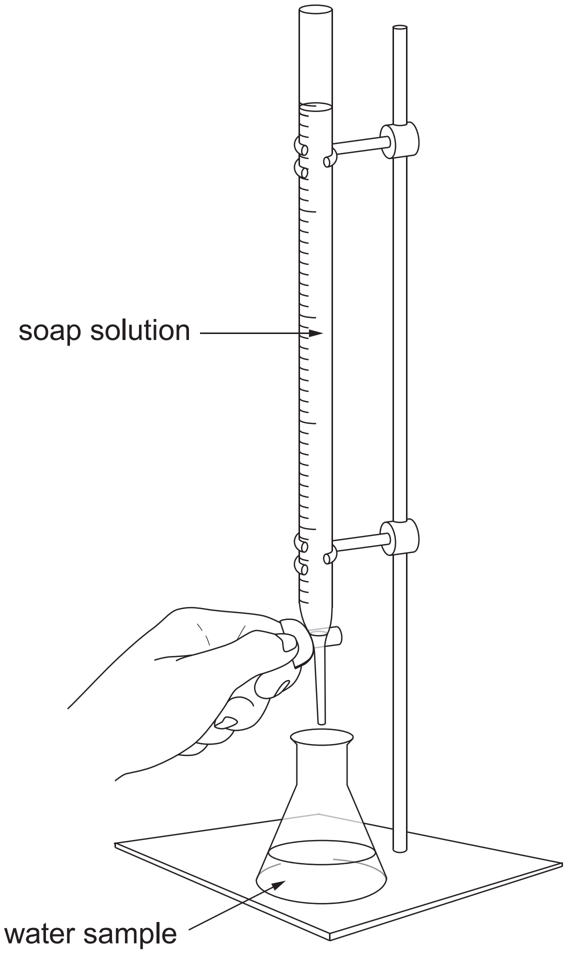
15

8

6. Olivia investigated the relative hardness of three samples of water, **A**, **B** and **C**.



She added soap solution to each sample separately and recorded the volume of soap solution needed to get a permanent lather. Her results are shown in the table.

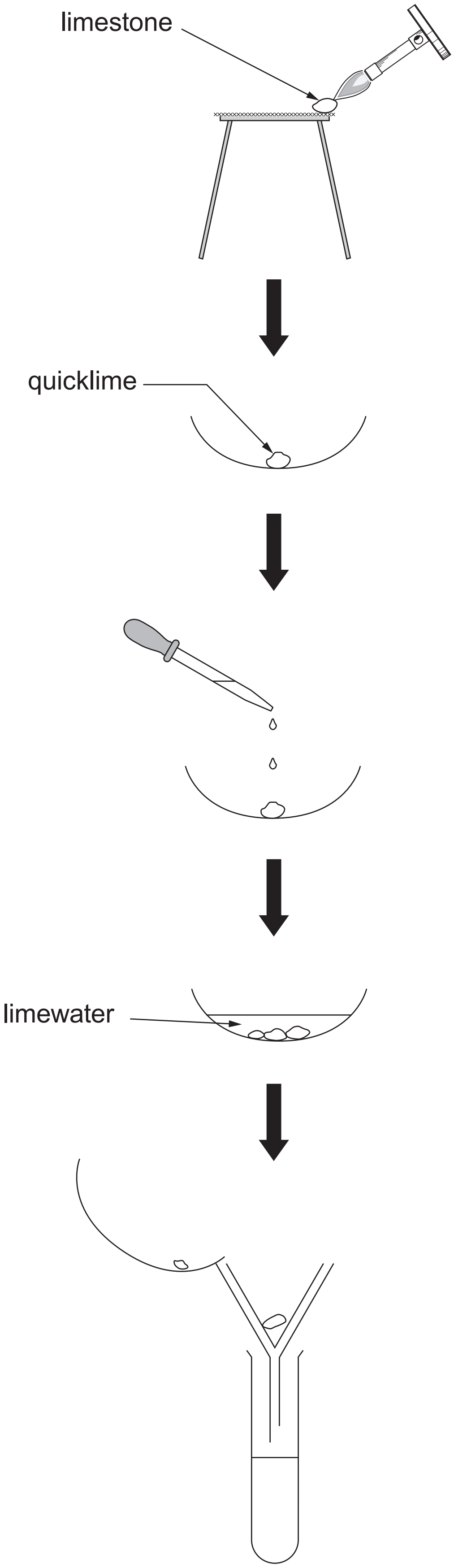
Water sample	Volume of soap solution needed to get a permanent lather (cm^3)
A	1.0
B	14.0
C	11.0

Examiner only

Describe how you would repeat this investigation, including enough detail to show that you will be carrying out a fair test. Assuming that you get similar results to Olivia, state and explain the conclusion you would reach. [6 QER]

6

7. The flow chart shows the stages a student carried out to change calcium carbonate (limestone) into calcium hydroxide solution (limewater).



Stage 1 Limestone is roasted for 15 minutes to make quicklime

Stage 2 Quicklime is cooled for 20 minutes

Stage 3 A few drops of water are added to the cold quicklime to make slaked lime

Stage 4 Excess water is added to the slaked lime to make limewater

Stage 5 Limewater is filtered

Examiner only

(a) Use the information in the flow chart to answer parts (i) to (iii).

(i) Name the type of reaction taking place in stage **1**. [1]

....

(ii) Give the **number** of the stage which shows an exothermic reaction. [1]

....

(iii) I. Describe what you would expect to **see** happen in stage **3**. [2]

....

....

....

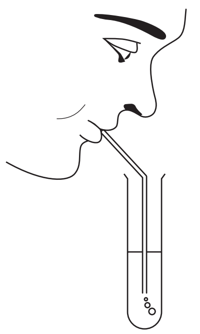
II. The reaction taking place in stage **3** is shown by the following word equation.

calcium oxide + water $\longrightarrow$ calcium hydroxide

Write a balanced **symbol** equation for this reaction. [2]

.... + ... $\longrightarrow$ ....

(b) The student blew gently through a straw placed in a test tube containing limewater.



Describe what you would expect to happen to the limewater. Give a reason for your answer. [2]

Observation ....

Reason ....

Examiner only

(c) Quicklime is manufactured from limestone in a lime kiln. A manufacturer expected to obtain 5.6 tonnes per 10 tonnes of limestone used. The actual mass of quicklime produced was 5.1 tonnes.

Calculate the percentage yield of quicklime for this manufacturing process. [2]

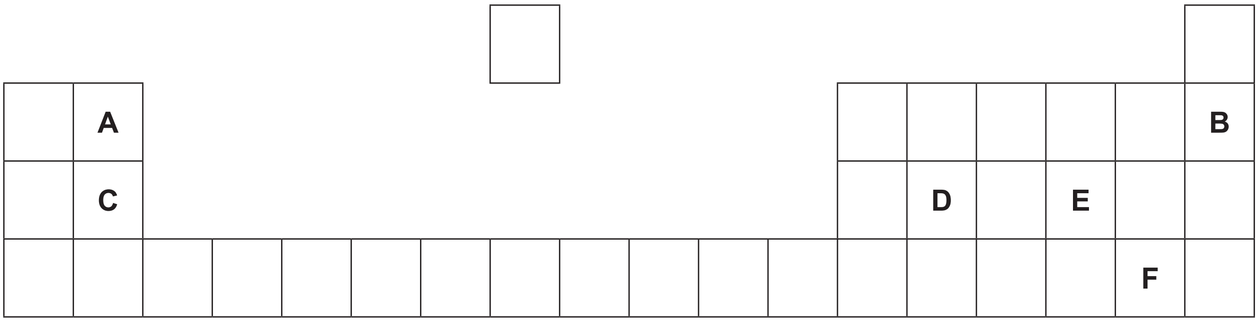
percentage yield = ... %

10

Examiner only

8. *(a)* The following diagram shows an outline of part of the Periodic Table.

The letters shown are NOT the chemical symbols of the elements.

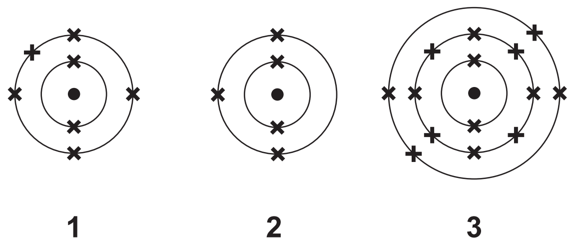


Choose **letters** from the diagram to complete the table below. [5]

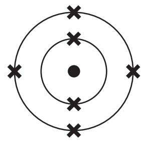
	Letter(s)
Two elements in the same group	.... and ....
The element which has 12 protons in its nucleus	....
The element with a full outer shell of electrons	....
The element in Group 2 and Period 2	....
The element with the electronic structure 2,8,4	....

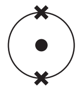
Examiner only

(b) Diagrams **1-5** show the electronic structure of five elements in the Periodic Table.



Give the **number** of the diagram which shows the electronic structure of the element which lies

(i) directly **below**  in the Periodic Table, ... [1]

(ii) to the **left** of  in the Periodic Table. .... [1]

(c) Nitrogen has two stable isotopes – nitrogen-14 and nitrogen-15.

Describe how these isotopes are similar to one another and how they are different. [2]

....

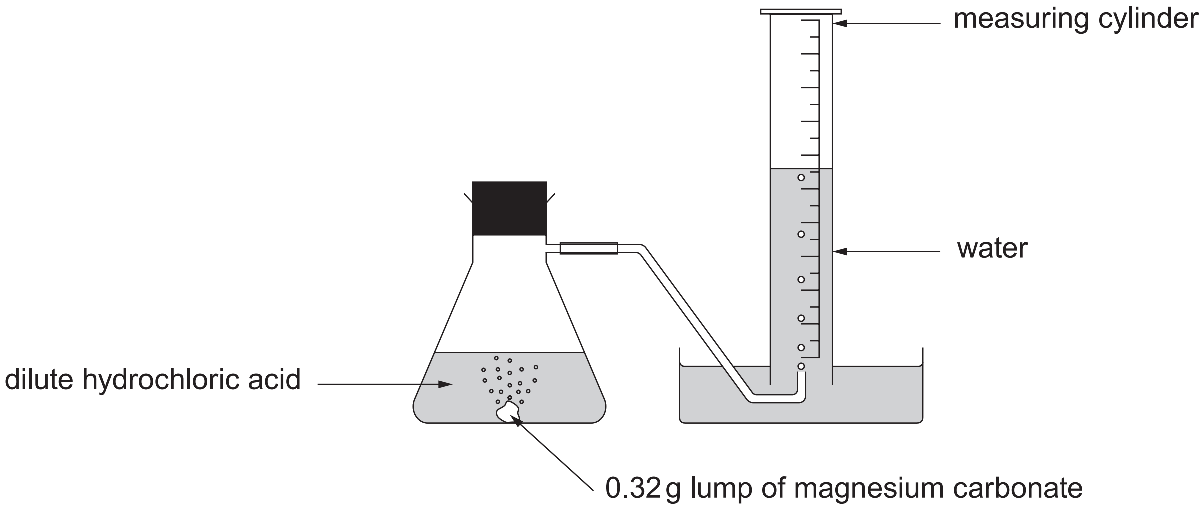
....

....

9

9. A student carried out an experiment to investigate the speed of the reaction between a **lump** of magnesium carbonate of mass 0.32 g and excess dilute hydrochloric acid at 20 °C.

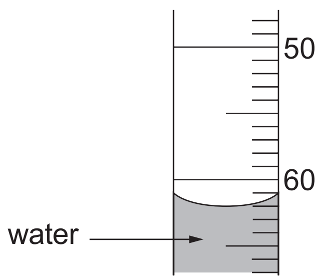
Magnesium carbonate reacts with dilute hydrochloric acid forming carbon dioxide gas. The total volume of carbon dioxide formed was recorded every 5 minutes for 40 minutes.



The results are shown below. The result for 15 minutes is missing.

Time (minutes)	0	5	10	15	20	25	30	35	40
Volume of carbon dioxide formed (cm^3)	0	20	41		79	83	90	90	90

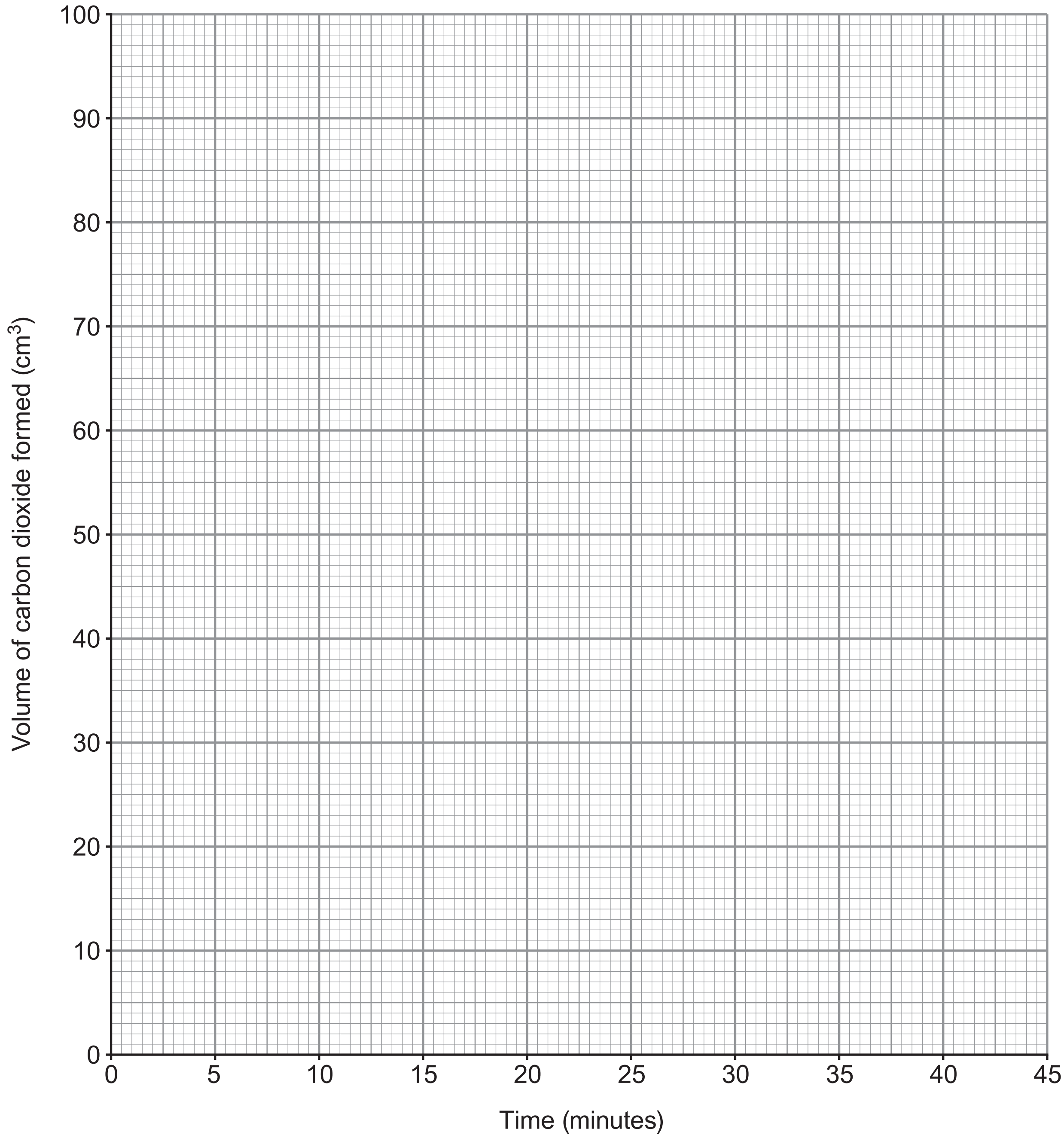
(a) Use the diagram below to find the volume of carbon dioxide gas formed after 15 minutes. [1]



volume of carbon dioxide = ... cm^3

Examiner only

(b) Plot the results from the table, including your answer to part *(a)*, on the grid below. Draw a suitable line and **label this graph X**. [3]



(c) Sketch the graph you would expect if the experiment were repeated using 0.32 g of magnesium carbonate **powder** instead of the lump of magnesium carbonate.
Label this graph Y. [2]

Examiner only

(d) State and explain, using particle theory, the effect of **increasing** the concentration of the hydrochloric acid. [3]

....

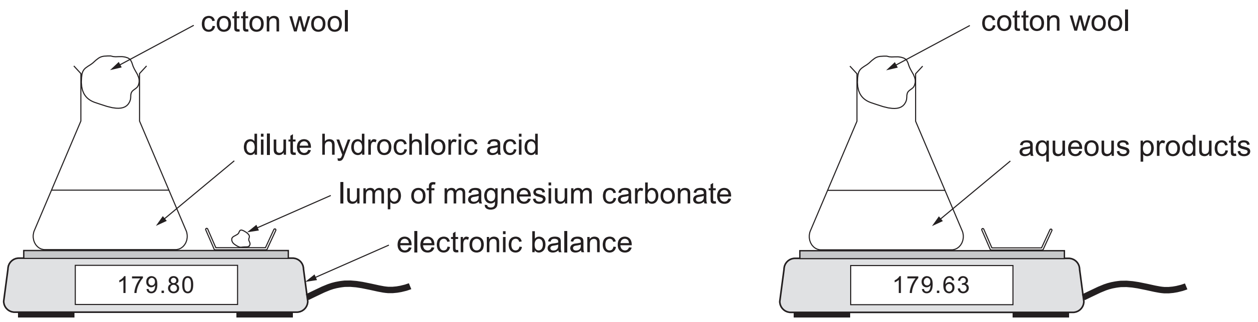
....

....

....

....

(e) The student investigated the same reaction using a different apparatus. A lump of magnesium carbonate was added to excess dilute hydrochloric acid at 20 °C.

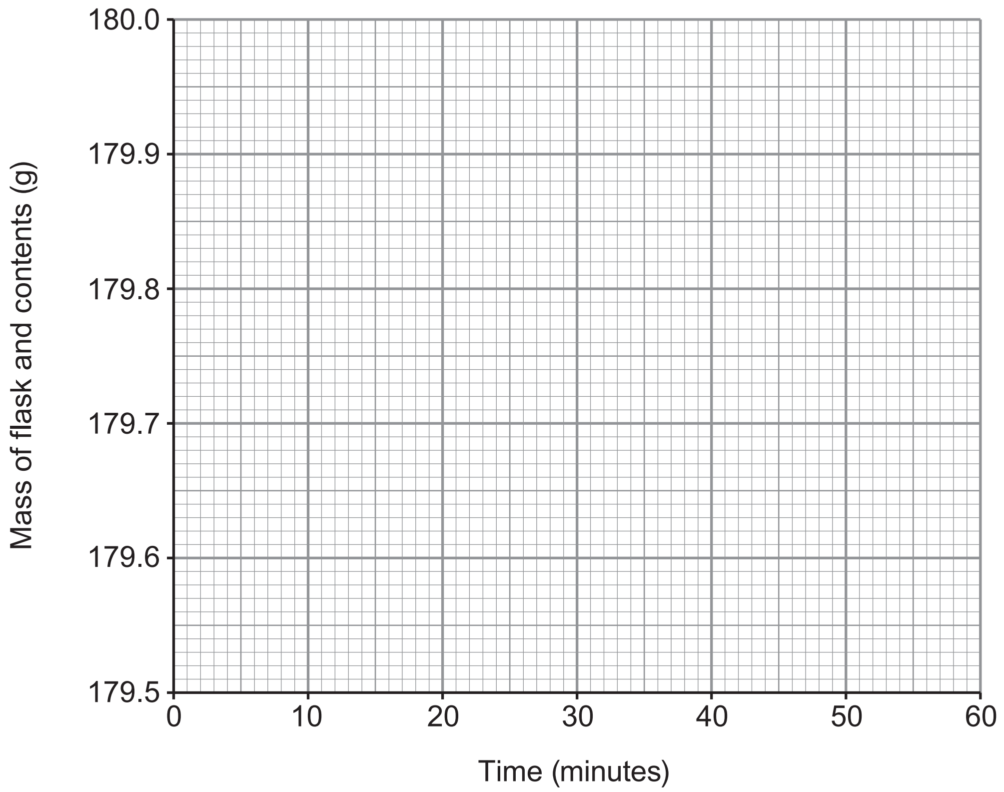


Apparatus at the start

Apparatus after 60 minutes

The change in mass was recorded for 60 minutes and displayed as a graph on a computer screen. The reaction took 40 minutes to complete.

Sketch the graph you would expect to see. [2]



Examiner only

11

END OF PAPER

Examiner only

For continuation only.

FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	**Formula**	**Name**	**Formula**
aluminium	Al^{3+}	bromide	Br^{-}
ammonium	NH_4^{+}	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^{-}
calcium	Ca^{2+}	fluoride	F^{-}
copper(II)	Cu^{2+}	hydroxide	OH^{-}
hydrogen	H^{+}	iodide	I^{-}
iron(II)	Fe^{2+}	nitrate	NO_3^{-}
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^{+}	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^{+}		
silver	Ag^{+}		
sodium	Na^{+}		
zinc	Zn^{2+}		

THE PERIODIC TABLE

1	2					Group						3	4	5	6	7	0
						1 H Hydrogen 1											4 He Helium 2
7 Li Lithium 3	9 Be Beryllium 4											11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10
23 Na Sodium 11	24 Mg Magnesium 12											27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulfur 16	35.5 Cl Chlorine 17	40 Ar Argon 18
39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36
86 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	99 Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54
133 Cs Caesium 55	137 Ba Barium 56	139 La Lanthanum 57	179 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86
223 Fr Francium 87	226 Ra Radium 88	227 Ac Actinium 89															

Key

A_r — relative atomic mass
Symbol
Name
Z — atomic number

Surname	Centre Number	Candidate Number
Other Names		0

GCSE – NEW

3410U10-1

S18-3410U10-1

CHEMISTRY – Unit 1:
Chemical Substances, Reactions and Essential Resources

FOUNDATION TIER

WEDNESDAY, 13 JUNE 2018 – MORNING

1 hour 45 minutes

For Examiner's use only		
Question	**Maximum Mark**	**Mark Awarded**
1.	**9**	
2.	**6**	
3.	**7**	
4.	**8**	
5.	**10**	
6.	**9**	
7.	**5**	
8.	**6**	
9.	**6**	
10.	**9**	
11.	**5**	
Total	**80**	

3410U101
01

ADDITIONAL MATERIALS

In addition to this examination paper you will need a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question **8** is a quality of extended response (QER) question where your writing skills will be assessed.

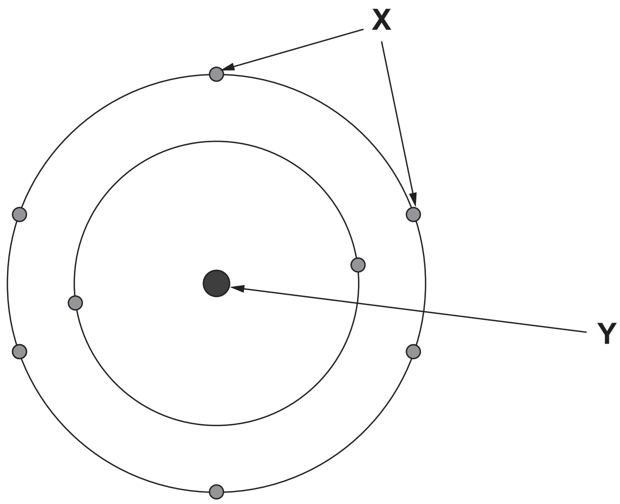
The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.

JUN183410U10101

Examiner only

Answer **all** *questions.*

1. *(a)* The following diagram shows the structure of an atom.



electrons	**protons**	**shells**	**neutrons**	**nucleus**

Complete the following sentences using words from the box. [3]

The particles labelled **X** are called ....

The part of the atom labelled **Y** is called the ....

It contains particles called .... and ....

(b) Draw a line from each particle to the charge of that particle. [2]

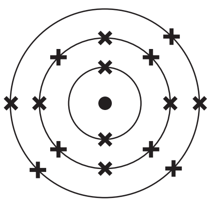
proton	**0**
neutron	**+1**
electron	**−1**

 (3410U10-1)

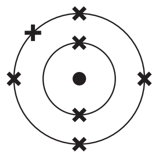
Examiner only

(c) The diagrams below show five different atoms labelled **A**-**E**.

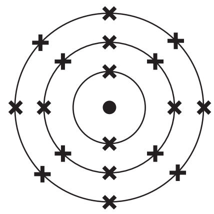
These letters are **not** the chemical symbols for the elements.



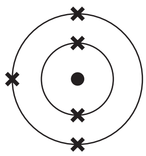
A



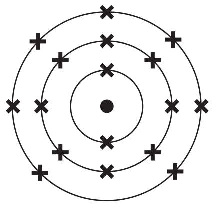
B



C



D



E

Use the letters **A**-**E** to complete the following sentences.

(i) The atoms found in Group 5 are .... and .... [1]

(ii) The atom with an atomic number of 7 is .... [1]

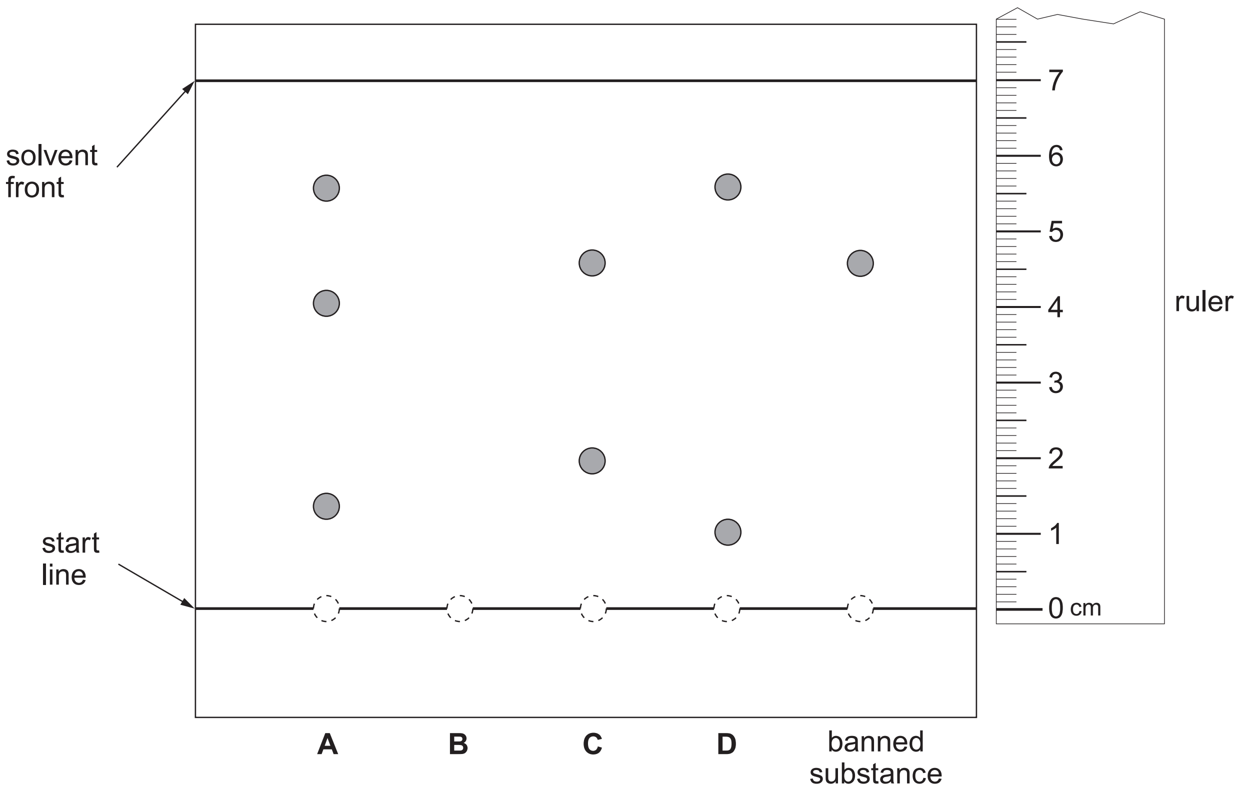
(iii) The atom that is an inert gas is .... [1]

(iv) The atoms found in Period 2 are .... and .... [1]

9

Examiner only

2. A food company was accused of using a banned substance in its sweets. Scientists tested four dyes, **A**-**D**, to find out if this was true or not. The results are shown below.



(a) Name the method used. [1]

....

(b) Dye **B** was found to have an R_f value of 0.428.

(i) Use the following equation to calculate the distance moved by dye **B**. [2]

distance moved by dye **B** = R_f × distance moved by solvent front

Distance moved = ... cm

(ii) **Complete the diagram** above to show the position of dye **B**. [1]

 (3410U10-1)

Examiner only

(c) Identify which dye, **A-D**, contains the banned substance. Give a reason for your answer. [2]

Dye ....

Reason ....

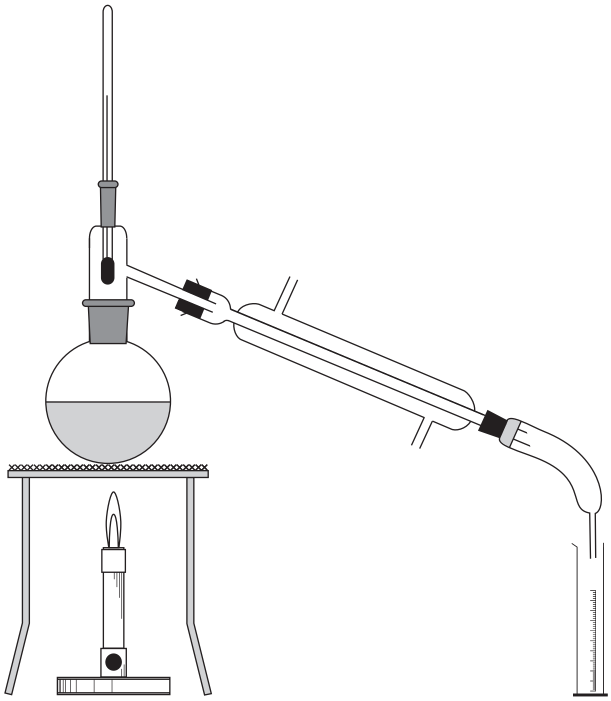
....

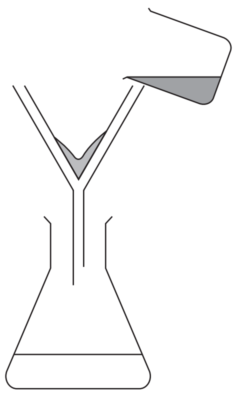
6

3410U101
05

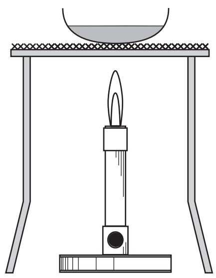
3. *(a)* The diagrams below show three methods of separating mixtures.

Method **A**





Method **B**



Method **C**

Examiner only

(i) A student was given a solution of sodium chloride. State which method, **A**, **B** or **C**, he could use in order to obtain a sample of pure water from the solution. Explain how the method works. [3]

Method ....

Explanation ....

....

....

(ii) 500 g of solution was found to contain 43 g of sodium chloride. Calculate the percentage of sodium chloride in the solution. [2]

Percentage = ... %

3410U101 07

(b) Group 7 ions, chloride, bromide and iodide, can be identified using silver nitrate solution.

Complete the following table to show the colours of the precipitates produced by these ions. [2]

Group 7 ion	Colour of precipitate
chloride	white
bromide	....
iodide	....

7

Examiner only

4. The following table shows the composition of the atmosphere.

Gas	Chemical formula	Percentage found in the atmosphere (%)
argon	Ar	0.93
carbon dioxide	CO_2	0.0360
helium	He	0.0005
hydrogen	H_2	0.00005
methane	CH_4	0.00017
neon	Ne	0.0018
nitrogen	N_2	78.08
nitrous oxide	N_2O	0.00003
oxygen	O_2	20.95
ozone	O_3	0.000004

Use the table to answer parts *(a)* and *(b)*.

(a) (i) Name **two** gases that occur as single atoms. [1]

....

(ii) Name **two** elements that occur as molecules. [1]

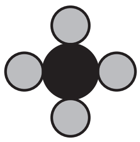
....

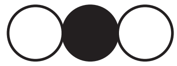
(iii) Name the gas that has the **lowest** percentage. [1]

....

Examiner only

(b) (i) If  represents a carbon atom, name the gases from the table that can be represented by the following diagrams. [2]





....

(ii) Use the information in part (i) to draw the diagram for oxygen gas. [1]

(c) The atmosphere can become polluted by substances known as chlorofluorocarbons (CFCs). One example of a CFC is CH_2ClF.

3410U101 09

(i) State how many hydrogen atoms are present in CH_2ClF. [1]

....

(ii) Give the **total** number of atoms shown in the formula, CH_2ClF. [1]

....

8

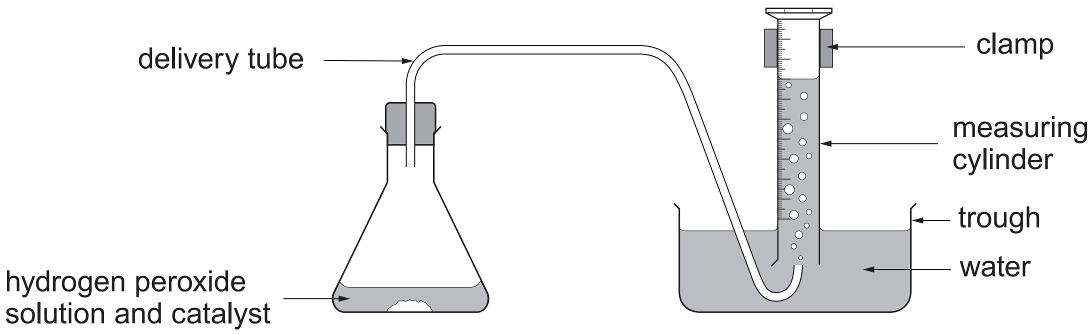
Examiner only

5. Hydrogen peroxide decomposes to give water and oxygen.

(a) Complete the symbol equation to show the reaction taking place. [2]

$$2H_2O_2 \longrightarrow \boxed{}\, H_2O \;+\; \ldots\ldots\ldots\ldots\ldots\ldots$$

(b) The rate of decomposition of hydrogen peroxide can be measured using the following apparatus.



The rate was investigated using three different catalysts. The results are shown in the table.

Time (s)	Volume of gas collected (cm^3)		
	Catalyst 1	Catalyst 2	Catalyst 3
0	0	0	0
20	2	20	8
40	4	34	15
60	6	38	23
80	8	40	30
100	10	40	36

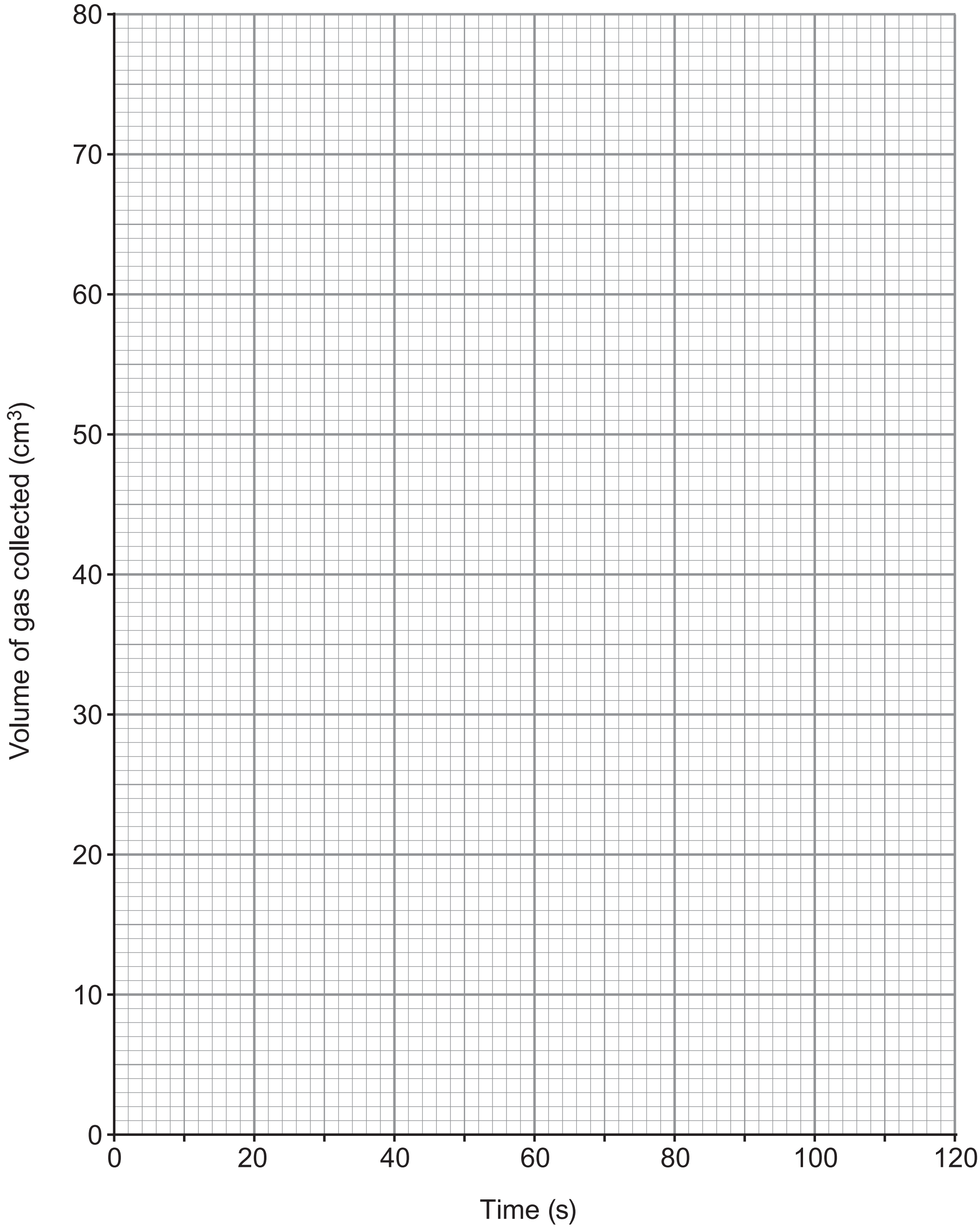
(i) State which is the **least** effective catalyst. Give a reason for your answer. [1]

....

....

Examiner only

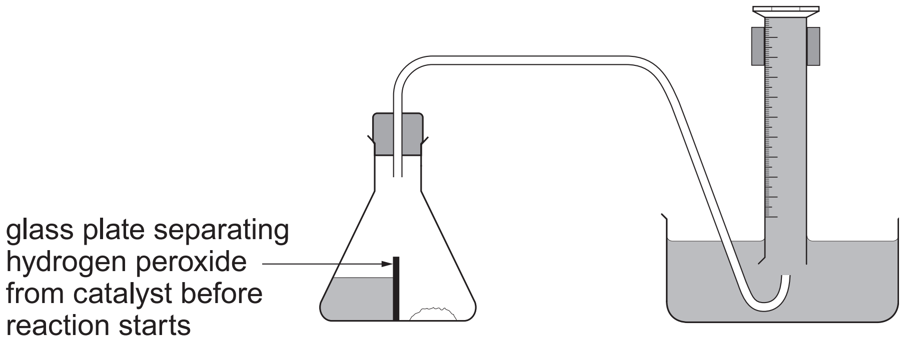
(ii) Plot a graph of the volume of gas collected using **catalyst 2**.
Draw a suitable line. [3]



(iii) **On the same grid**, sketch the graph you would expect to obtain if you added the same amount of catalyst 2 to the same volume of hydrogen peroxide of **twice** the concentration. [2]

Examiner only

(iv) Another student claimed that he could collect more accurate results using the following apparatus.



Suggest how this apparatus could improve the accuracy of the results. [2]

....

....

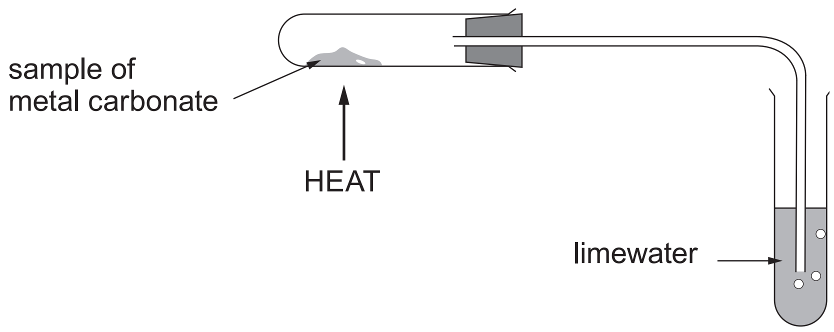
....

10

Examiner only

6. *(a)* A student investigated the decomposition of three different metal carbonates.

She measured the time taken for limewater to turn milky using the following apparatus.



Three samples of each metal carbonate were tested. Her results are shown in the table.

Metal carbonate	Time taken for limewater to turn milky (s)			
	Sample 1	Sample 2	Sample 3	Mean
copper(II) carbonate	15	25	17	....
zinc carbonate	54	52	53	53
calcium carbonate	195	200	190	195

(i) Calculate the mean time taken for limewater to turn milky on heating copper(II) carbonate. **Show your working.** [2]

Mean time = ... s

Examiner only

(ii) I. Place the carbonates in order of stability giving a reason for your answer. [2]

Most stable ....

....

Least stable ....

Reason ....

....

II. Explain the order of stability of the carbonates. [1]

....

....

(iii) Complete the following symbol equation for the decomposition of copper(II) carbonate. [2]

$CuCO_3 \longrightarrow$ .... + ...

(b) Calculate the relative formula mass, M_r, of copper(II) carbonate, $CuCO_3$. [2]

$A_r(C) = 12$ $A_r(O) = 16$ $A_r(Cu) = 63.5$

M_r = ...

9

Examiner only

7. The following information is taken from some articles about global warming.

Greenhouse gases such as carbon dioxide keep heat close to the Earth's surface making it a suitable temperature for life. Global warming is an increase of the Earth's mean surface temperature due to the overproduction of greenhouse gases by burning fossil fuels such as gas, petrol and oil. Deforestation also contributes to this. With the growth of industry in the 1900s, humans began burning more fossil fuels to run our cars, trucks and factories. There is more carbon dioxide in the atmosphere today than at any point in the last 800,000 years.

The following charts show the mean global temperature every decade since the 1880s, the amount of carbon dioxide in the atmosphere from 1750-2010 and the main sources of carbon dioxide production today.

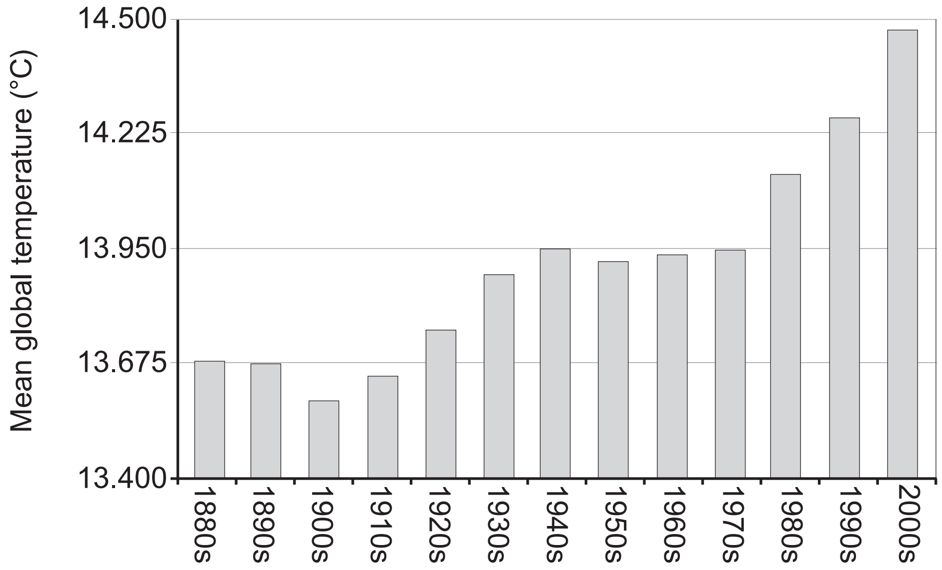


Figure 1

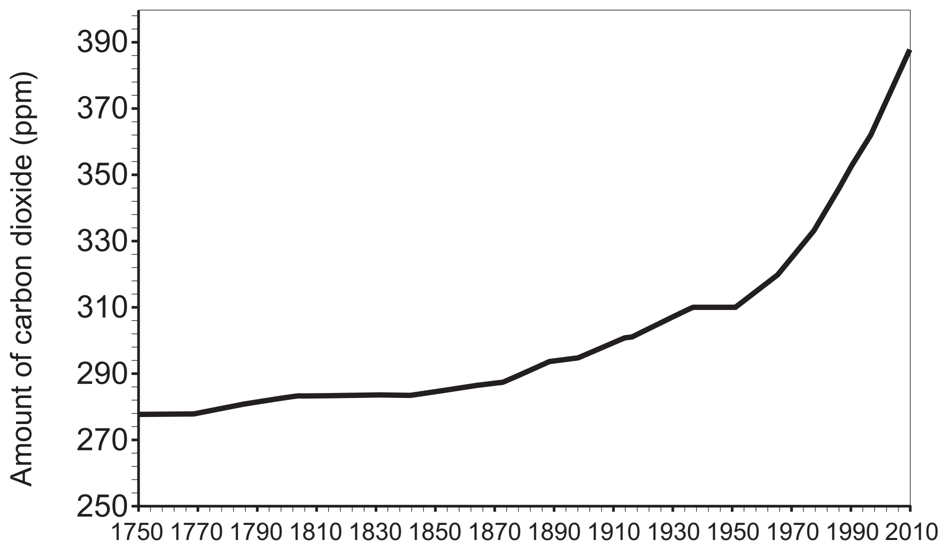


Figure 2

Examiner only

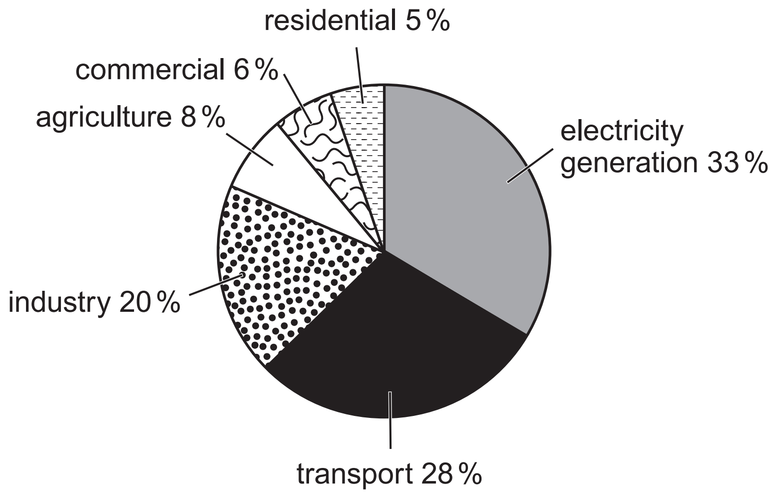


Figure 3

(a) Draw a trend line on **Figure 1**. [1]

(b) Suggest why there was a large rise in the amount of carbon dioxide in the atmosphere after around 1950. [2]

....

....

....

....

(c) Give **one** way in which we are trying to reduce carbon dioxide emissions. [1]

....

....

(d) Some people say that changes in carbon dioxide levels are **not** responsible for global warming. Tick (✓) the statement that supports this opinion. [1]

Most carbon dioxide is produced by electricity generation ☐

Between 1900 and 2010 there was a massive increase in industry ☐

The mean global temperature remained constant between 1950 and 1980 ☐

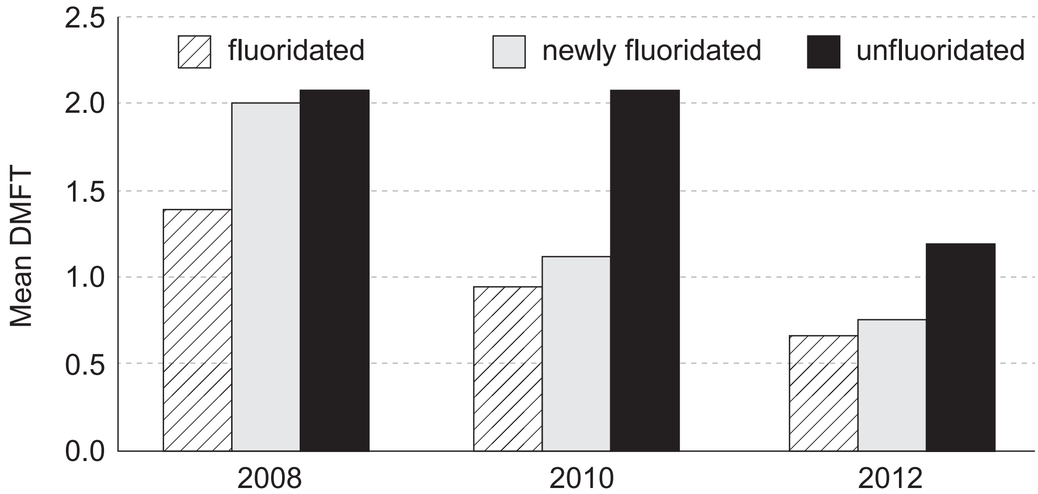
The average number of cars per home has increased steadily since the 1980s ☐

The use of energy efficient appliances has increased since 2000 ☐

5

Examiner only

8. The following graph shows the mean numbers of decayed, missing and filled teeth (DMFT) in 12 year-olds in three areas of Australia in 2008, 2010 and 2012. One area has been fluoridated for 20 years, one is newly fluoridated and the other is unfluoridated.



Describe what the graph tells us. Use this information and your knowledge of fluoridation to explain why some people support fluoridation of water supplies but others oppose it. [6 QER]

....

....

....

....

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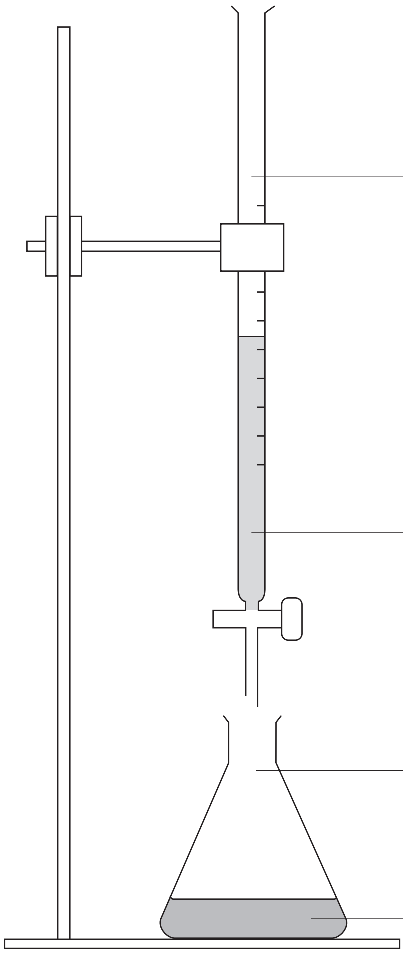
6

9. Water samples **A**, **B**, **C** and **D** were tested for hardness using the apparatus shown.



Soap solution was added 1 cm^3 at a time to each sample and the volume required to produce a permanent lather on shaking was recorded. Each sample was tested before and after boiling. The results are shown in the table.

Water sample	Volume of soap solution required (cm^3)	
	Before boiling	After boiling
A	1	1
B	10	10
C	15	1
D	15	8

Examiner only

(a) (i) State which water sample contains **only** temporary hardness. Explain your answer. [2]

Water sample ....

Explanation ....

....

....

(ii) Give **one** similarity in the composition of temporary and permanent hard water. [1]

....

....

(b) Discuss the benefits and drawbacks of living in a hard water area. [3]

....

....

....

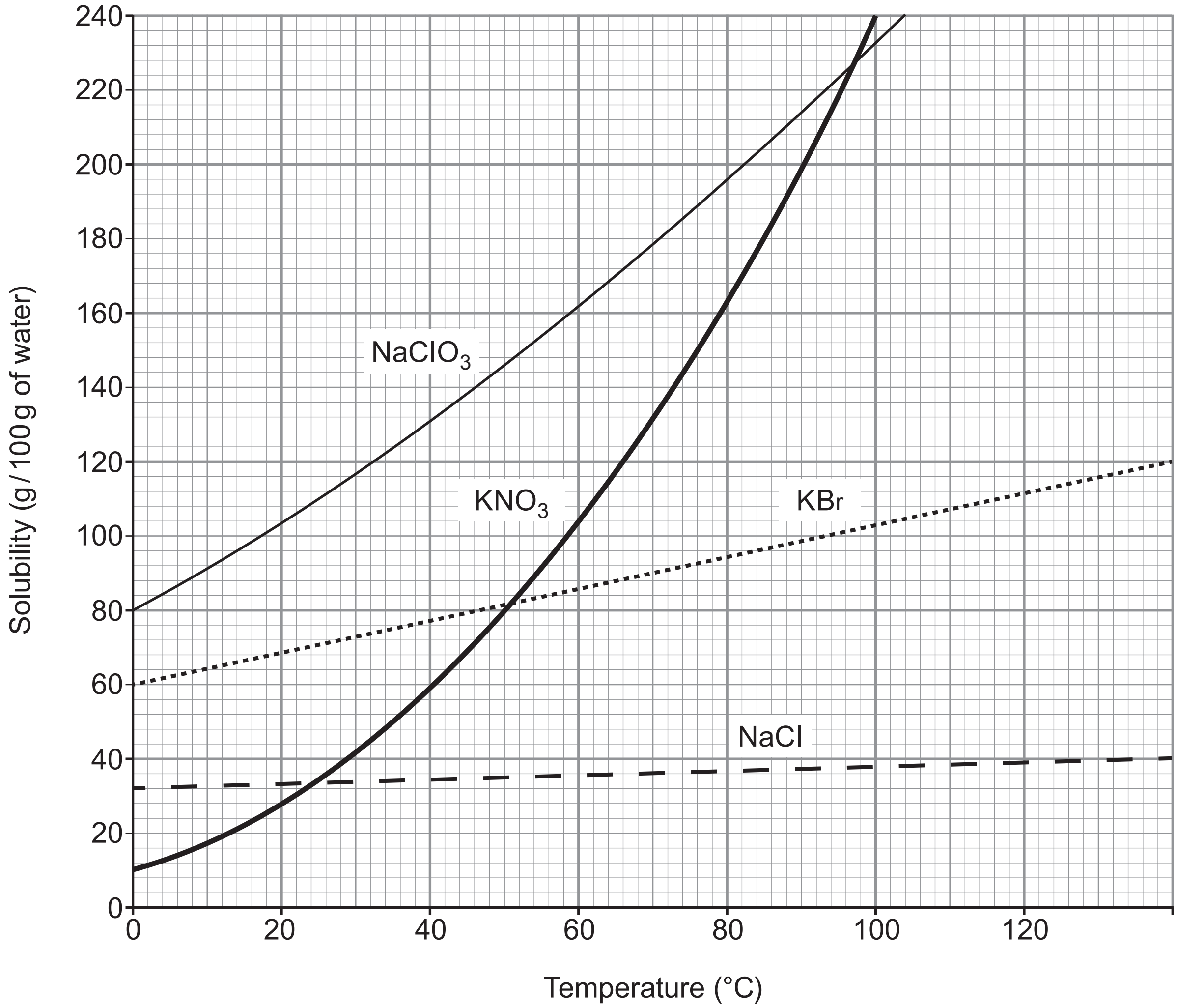
....

....

....

6

10. The grid below shows the solubility curves for four ionic compounds.



$NaClO_3$	sodium chlorate
KNO_3	potassium nitrate
KBr	potassium bromide
NaCl	sodium chloride

Examiner only

(a) (i) Give the temperature at which the solubility of potassium nitrate and potassium bromide is the same. [1]

.... °C

(ii) Calculate the mass of solid potassium nitrate that would form if a saturated solution in 200 g of water were cooled from 100 °C to 20 °C. [3]

Mass = ... g

(iii) Suggest why a student may be surprised at the temperature range shown on the solubility curves. [1]

....

....

(b) (i) Give the symbols of the **ions** of Group 1 elements present in the compounds shown on the grid. [1]

....

(ii) Explain how these ions are formed from their atoms. [2]

....

....

(c) Potassium nitrate reacts with aluminium hydroxide to produce aluminium nitrate and potassium hydroxide.

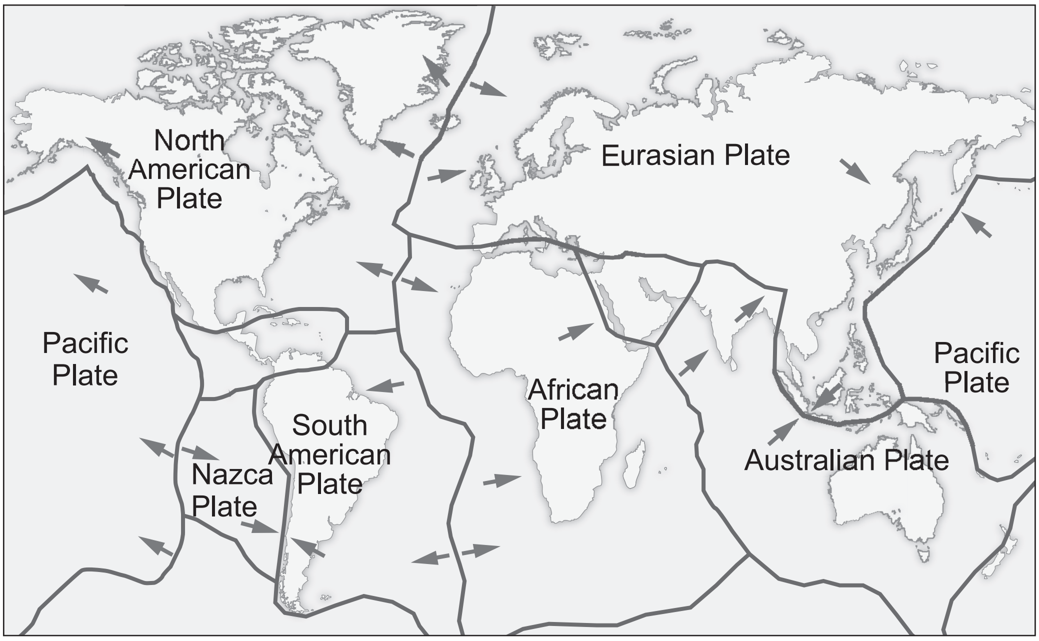
Balance the symbol equation for the reaction taking place. [1]

$$\boxed{}\ KNO_3 + Al(OH)_3 \longrightarrow Al(NO_3)_3 + \boxed{}\ KOH$$

9

Examiner only

11. The following diagram shows some of the Earth's tectonic plates and the direction in which they move.



(a) The boundary between the Nazca and South American plates is a destructive plate boundary. Describe what happens at a destructive boundary. [2]

....

....

....

(b) Draw a cross (**×**) on the diagram to show a constructive plate boundary.
Describe what happens at this boundary. [2]

....

....

....

(c) State **one** effect of plates sliding past each other. [1]

....

5

END OF PAPER

FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	**Formula**	**Name**	**Formula**
aluminium	Al^{3+}	bromide	Br^{-}
ammonium	NH_4^{+}	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^{-}
calcium	Ca^{2+}	fluoride	F^{-}
copper(II)	Cu^{2+}	hydroxide	OH^{-}
hydrogen	H^{+}	iodide	I^{-}
iron(II)	Fe^{2+}	nitrate	NO_3^{-}
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^{+}	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^{+}		
silver	Ag^{+}		
sodium	Na^{+}		
zinc	Zn^{2+}		

THE PERIODIC TABLE

1	2						Group					3	4	5	6	7	0
							1 H Hydrogen 1										4 He Helium 2
7 Li Lithium 3	9 Be Beryllium 4											11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10
23 Na Sodium 11	24 Mg Magnesium 12											27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulfur 16	35.5 Cl Chlorine 17	40 Ar Argon 18
39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36
86 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	99 Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54
133 Cs Caesium 55	137 Ba Barium 56	139 La Lanthanum 57	179 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86
223 Fr Francium 87	226 Ra Radium 88	227 Ac Actinium 89															

Key

A_r — relative atomic mass
Symbol
Name
Z — atomic number

Surname	Centre Number	Candidate Number
Other Names		0

GCSE

3410U10-1

S19-3410U10-1

WEDNESDAY, 12 JUNE 2019 – MORNING

CHEMISTRY – Unit 1:
Chemical Substances, Reactions and Essential Resources

FOUNDATION TIER

1 hour 45 minutes

For Examiner's use only		
Question	**Maximum Mark**	**Mark Awarded**
1.	**7**	
2.	**7**	
3.	**8**	
4.	**6**	
5.	**6**	
6.	**6**	
7.	**10**	
8.	**10**	
9.	**7**	
10.	**7**	
11.	**6**	
Total	**80**	

3410U101
01

ADDITIONAL MATERIALS

In addition to this examination paper you will need a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question **7***(a)* is a quality of extended response (QER) question where your writing skills will be assessed.

The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.

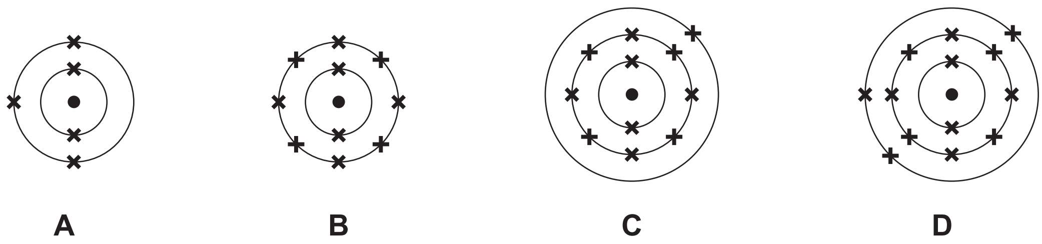
JUN193410U10101

Examiner only

Answer **all** *questions*.

1. *(a)* The diagrams show the atoms of four elements, **A**-**D**.

These letters are **not** the chemical symbols for the elements.



(i) Use the letters **A**-**D** to complete the following sentences. [2]

Two atoms found in the same group are .... and ....

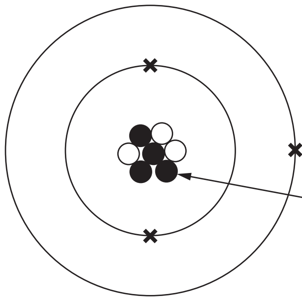
The atom found in Group 1 is ....

(ii) I Give the electronic structure of atom **B**. .... [1]

II Give the atomic number of atom **B**. .... [1]

Examiner only

(b) The following diagram shows an atom of lithium.



(i) Complete the following sentences. [2]

Particle **X** is an ....

Particle **Y** found in the nucleus is a ....

(ii) Use the diagram to explain why lithium has a mass number of 7. [1]

....

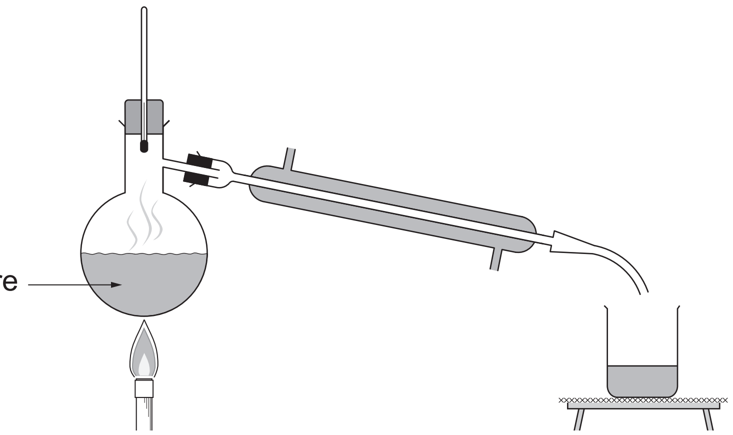
....

3410U101
03

7

Examiner only

2. *(a)* A teacher carried out an experiment to show how to separate a mixture of ethanol and water.



(i) Choose from the box the name of the method used. [1]

chromatography	**filtration**	**distillation**	**evaporation**

Method ....

(ii) The following sentences show the steps involved in this method of separation.

A Ethanol vapour cools and condenses
B Ethanol boils and evaporates
C The mixture of water and ethanol is heated
D Ethanol is collected

Put the steps in the correct order. Use the letters **A-D**. [1]

....

(iii) Tick (✓) the box below that explains why it is the water that is left in the flask. [1]

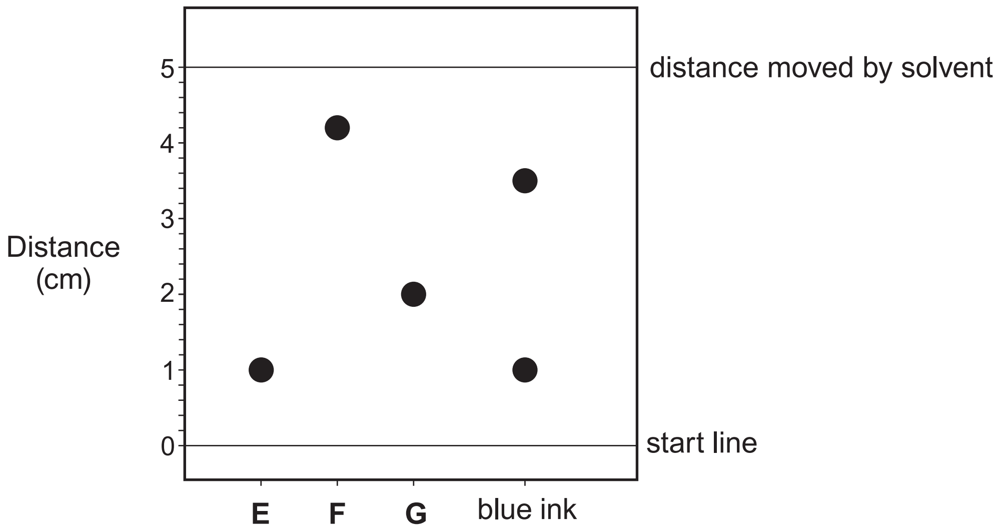
the boiling point of ethanol is equal to the boiling point of water ☐

the boiling point of ethanol is lower than the boiling point of water ☐

the boiling point of ethanol is higher than the boiling point of water ☐

Examiner only

(b) A student carried out an investigation to find out whether pigments **E**, **F** and **G** are present in a blue felt pen. The results are shown below.



(i) State what the results tell you about the blue ink. [2]

....

....

(ii) Calculate the R_f value of pigment **F** using the equation given. [2]

$$R_f = \frac{\text{distance moved by pigment}}{\text{distance moved by solvent}}$$

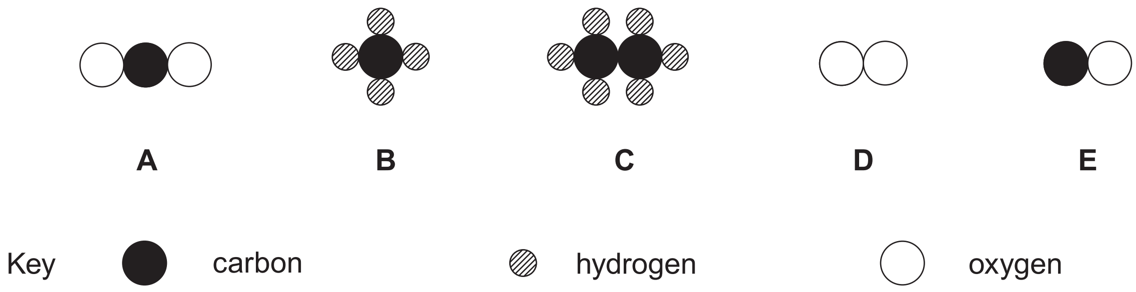
R_f = ...

7

3410U101 05

Examiner only

3. *(a)* The diagrams represent five substances, **A-E**.



(i) State which diagram, **A-E**, represents an element. Give a reason for your answer. [2]

Element ....

Reason ....

....

(ii) State which diagram, **A-E**, represents carbon dioxide, CO_2. [1]

....

(iii) Give the formula of the substance represented by diagram **C**. [1]

....

(b) When copper(II) sulfate solution reacts with sodium hydroxide, it forms copper(II) hydroxide, $Cu(OH)_2$, and sodium sulfate, Na_2SO_4.

(i) Write a **word** equation for the reaction taking place. [1]

....

(ii) Give the number of atoms of sulfur found in the formula Na_2SO_4. [1]

....

(iii) Give the **total** number of atoms in the formula $Cu(OH)_2$. [1]

....

(c) Iron(III) sulfate reacts with sodium hydroxide to produce iron(III) hydroxide.

Write the formula of iron(III) hydroxide. [1]

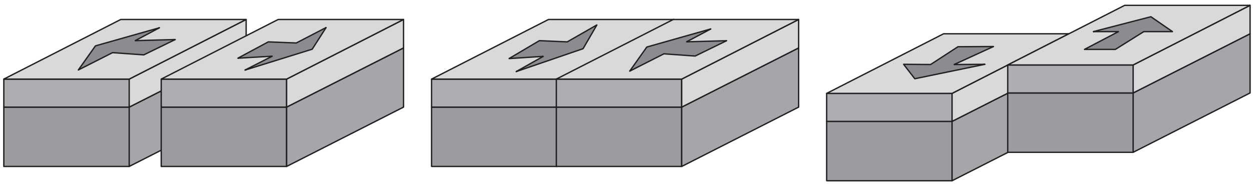
....

8

Examiner only

4. The surface of the Earth is divided into a number of tectonic plates. These plates are constantly moving.

Plate boundaries can be classified according to the direction of movement of the plates. The three main types of boundary are shown below.



A **B** **C**

conservative	**destructive**	**constructive**

(a) Choose the name of each type of boundary from the box. [2]

A ....

B ....

C ....

(b) Describe what happens as a result of the movement shown at boundary **A**. [2]

....

....

....

(c) Explain what happens at boundary **C** as a result of the plate movement shown. [2]

....

....

....

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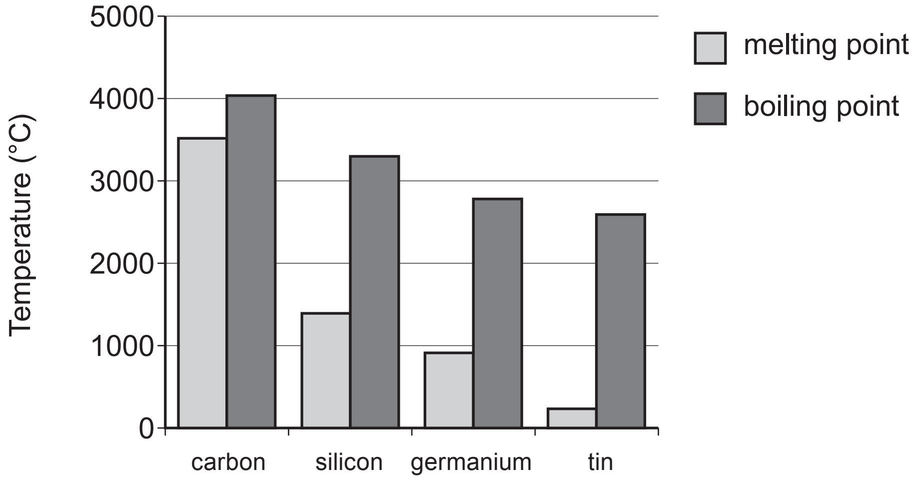
07

6

Examiner only

5. *(a)* The chart below shows the melting points and boiling points of some Group 4 elements.



(i) Describe the trend in melting point going down Group 4. [1]

....

(ii) Name the element which has the greatest **difference** between its melting point and boiling point. [1]

....

(b) (i) Name an element in Group 4 that is considered to be a semi-metal (metalloid). [1]

....

(ii) Tick (✓) the box below that describes where semi-metals are found in the Periodic Table. [1]

some are found in all groups ☐

they are all found in Group 4 ☐

some are found in Group 1 and Group 2 ☐

they are all found between metals and non-metals ☐

Examiner only

(c) Carbon is the main element found in coal. When carbon burns it forms carbon dioxide.

(i) Write a **word** equation to show the reaction taking place. [1]

....

(ii) Name **one** environmental problem caused by increased levels of carbon dioxide in the atmosphere. [1]

....

6

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09

6. This question is about the development of the Periodic Table.

In 1890, not all the elements had been discovered. A Russian scientist named Dmitri Mendeleev used the reactions of the known elements and their relative atomic masses to arrange them in the following table.

I	II	III	IV	V	VI	VII	VIII		
H 1.01									
Li 6.94	Be 9.01	B 10.8	C 12.0	N 14.0	O 16.0	F 19.0			
Na 23.0	Mg 24.3	Al 27.0	Si 28.1	P 31.0	S 32.1	Cl 35.5			
K 39.1	Ca 40.1		Ti 47.9	V 50.9	Cr 52.0	Mn 54.9	Fe 55.9	Co 58.9	Ni 58.7
Cu 63.5	Zn 65.4			As 74.9	Se 79.0	Br 79.9			
Rb 85.5	Sr 87.6	Y 88.9	Zr 91.2	Nb 92.9	Mo 95.9		Ru 101	Rh 103	Pd 106
Ag 108	Cd 112	In 115	Sn 119	Sb 122	Te 128	I 127			
Ce 133	Ba 137	La 139		Ta 181	W 184		Os 194	Ir 192	Pt 195
Au 197	Hg 201	Ti 204	Pb 207	Bi 209					
			Th 232		U 238				

Mendeleev's table

Based on the properties of the known elements, he predicted that new elements would be discovered and left gaps for these in his table. He even predicted what the properties of these undiscovered elements would be. He gave the name ekasilicon to one of them. The element that fits into this gap was eventually discovered and named germanium.

	Predicted properties of ekasilicon (Ek)	Actual properties of germanium (Ge)
Atomic mass	72	72.59
Density (g/cm^3)	5.5	5.35
Melting point	high	937.4 °C
Colour of metal	grey	grey-white
Formula of oxide	EkO_2	GeO_2
Formula of chloride	$EkCl_4$	$GeCl_4$

Comparison of the properties of ekasilicon and germanium

The modern day Periodic Table is based on Mendeleev's original table.

Examiner only

(a) (i) Put a tick (✓) in the correct column to show whether the following statements apply to Mendeleev's table only, today's table only or to both tables. [2]

	Mendeleev only	Today only	Both tables
the table is organised into groups			
copper and potassium are in the same group			
there are gaps in the table			
fluorine and chlorine are in the same group			

(ii) Tick (✓) the **two** statements that **best** describe why germanium was confirmed to be the element ekasilicon predicted by Mendeleev. [1]

germanium has exactly the same atomic mass as that predicted for ekasilicon ☐

germanium has a different colour to that predicted for ekasilicon ☐

germanium has a similar density to that predicted for ekasilicon ☐

germanium oxide has the same ratio of atoms as that predicted for ekasilicon oxide ☐

germanium oxide and germanium chloride have the same ratio of atoms ☐

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Examiner only

(b) (i) Calculate the percentage of oxygen present in the formula of germanium oxide, GeO_2. [2]

$A_r(Ge) = 73$ $A_r(O) = 16$

Percentage = ... %

(ii) When germanium oxide reacts with hydrochloric acid it produces germanium chloride and water.

Balance the following equation for the reaction that takes place. [1]

$$GeO_2 + \boxed{}\,HCl \longrightarrow GeCl_4 + \boxed{}\,H_2O$$

6

Examiner only

7. *(a)* The water used in our homes is treated to make it safe to drink.

State the main stages involved in the treatment of the public water supply. Describe the purpose of each stage. [6 QER]

You do not need to refer to fluoridation in your answer.

Examiner only

(b) Fluoride is sometimes added to water supplies.

State why fluoride is added and give **one** reason why some people oppose the addition of fluoride to water. [2]

Reason for adding fluoride

....

....

Reason why some people oppose its addition

....

....

(c) This advert was published to encourage people to save water by taking a shower instead of a bath.

Use the data in the advert to calculate the percentage of water **saved** by taking a shower for 8 minutes rather than running a full bath. [2]

Percentage of water saved = ... %

10

Examiner only

8. *(a)* A student was investigating how the concentration of sodium thiosulfate solution affects its reaction with hydrochloric acid.

(i) The reaction taking place can be represented by the following equation, which is **not** balanced.

Write the number **2** in **one of the boxes** on the right hand side in order to balance the equation. [1]

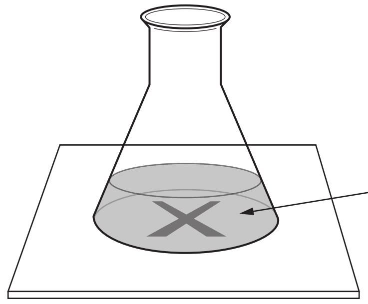
$$2HCl(aq) + Na_2S_2O_3(aq) \longrightarrow \boxed{}\ SO_2(g) + \boxed{}\ S(s) + \boxed{}\ NaCl(aq) + \boxed{}\ H_2O(l)$$

(ii) During the reaction, the solution becomes cloudy due to the formation of a precipitate. State the meaning of the term *precipitate*. [1]

....

....

(iii) The time taken for the cross to disappear was measured as shown below.

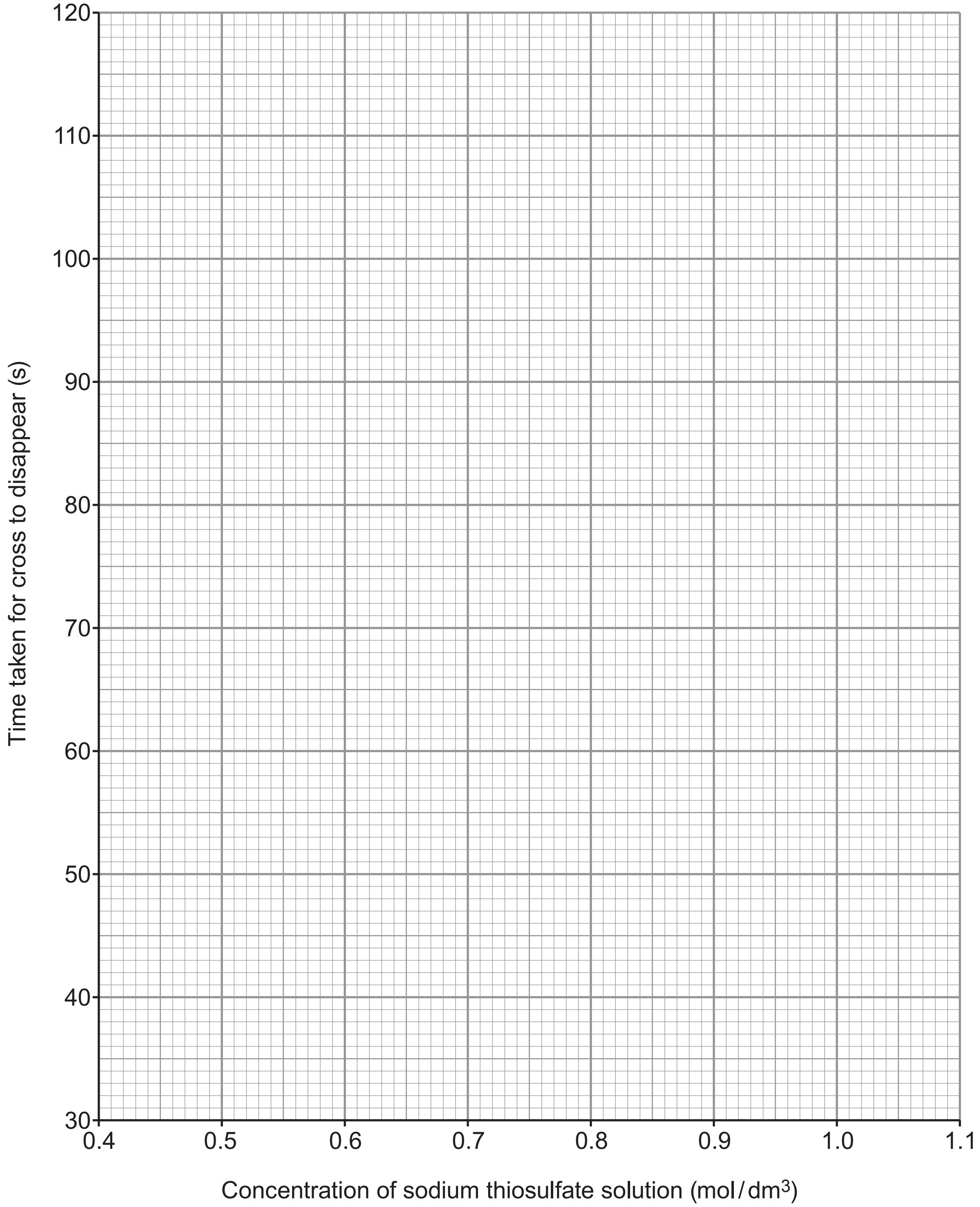


The results are shown in the table.

Concentration of sodium thiosulfate solution (mol / dm^3)	Time taken for cross to disappear (s)
1.0	42
0.9	46
0.8	53
0.7	66
0.6	87
0.5	110

Examiner only

Plot the results from the table on the grid. Draw a suitable line. [3]



Examiner only

(iv) State how concentration affects the **time taken** for the cross to disappear. [1]

....

....

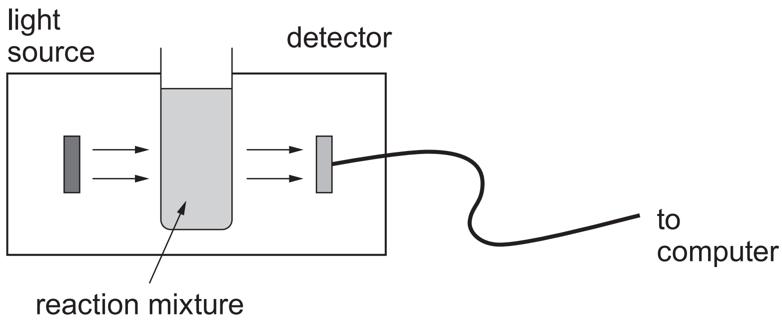
(v) What conclusion can be drawn about the effect of concentration on the **rate** of this reaction? [1]

....

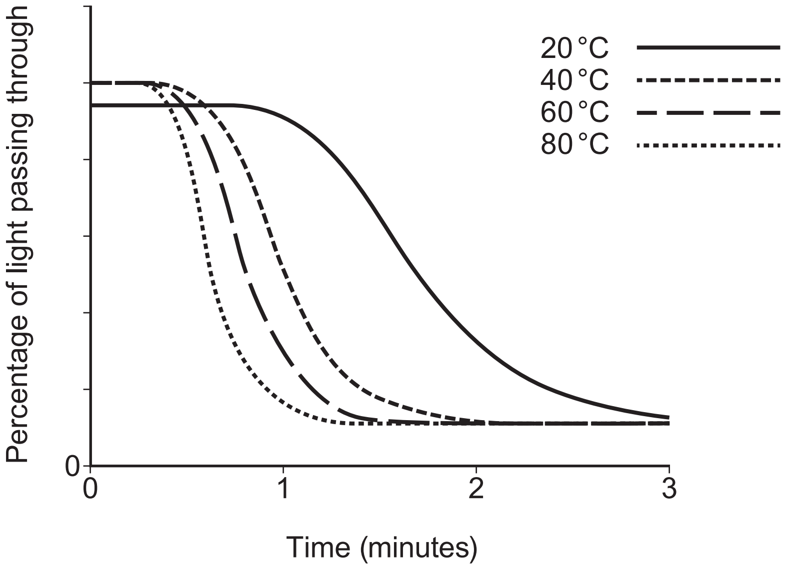
....

Examiner only

(b) A second student used a light sensor to investigate the effect of temperature on the rate of this reaction.



He obtained the following results.



(i) Give **one** conclusion that can be drawn from these results. State how this is shown by the graph. [2]

....

....

....

(ii) Slightly less light passed through the tube at the start of the experiment at 20 °C than in the others. Suggest a possible practical reason for this. [1]

....

....

10

Examiner only

9. *(a)* The following table shows some information about Group 1 elements.

Metal	Melting point (°C)	Boiling point (°C)	Density (g/cm^3)	Reaction with chlorine
lithium	180	1342	0.54	reacts slowly to make a white salt
sodium	97	883	0.97	burns vigorously with a yellow flame to make a white salt
potassium	63	759	0.88	reacts violently to make a white salt
rubidium	39	688	1.53	explosive reaction
caesium	28	671	1.93	explosive reaction

(i) Describe the trend in density going down the group. [1]

....

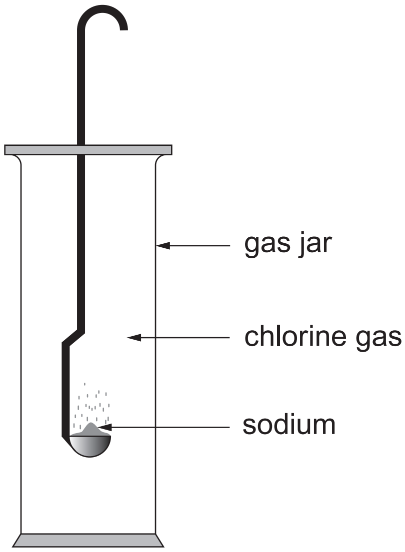
(ii) Explain the difference in reactivity down the group in terms of electronic structure. [2]

....

....

Examiner only

(b) The apparatus below can be used to demonstrate the reaction between sodium and chlorine.



(i) Apart from the use of safety goggles, state **one** safety precaution that needs to be followed when using **each** of these elements. [2]

Element	Safety precaution
sodium	…………………………………………
chlorine	…………………………………………

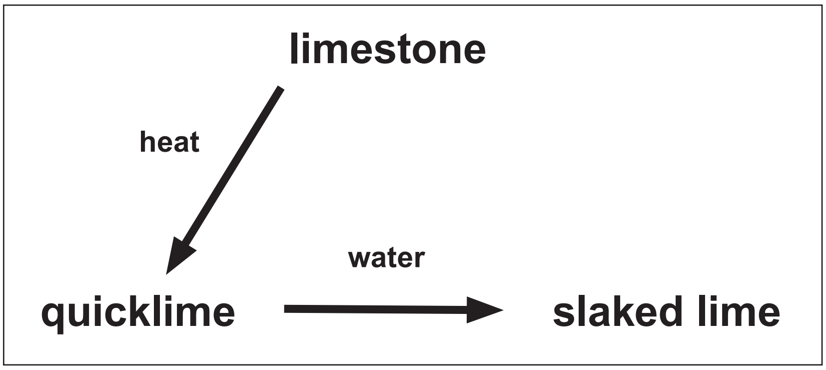
(ii) Complete and balance the symbol equation for the reaction that takes place between sodium and chlorine. [2]

$\square$ Na + Cl_2 $\longrightarrow$ $\square$ ………………………

7

Examiner only

10. The following diagram shows how limestone, $CaCO_3$, can be converted into useful products.



(a) When a piece of limestone is heated strongly its mass decreases.

State the type of reaction taking place. [1]

....

(b) (i) Describe what is **seen** when limestone is heated and converted into quicklime. [1]

....

(ii) Write a balanced symbol equation for the reaction taking place. [2]

....

(c) (i) Describe what is observed when quicklime is converted into slaked lime. [1]

....

(ii) Write a balanced symbol equation for the reaction taking place. [2]

....

7

Examiner only

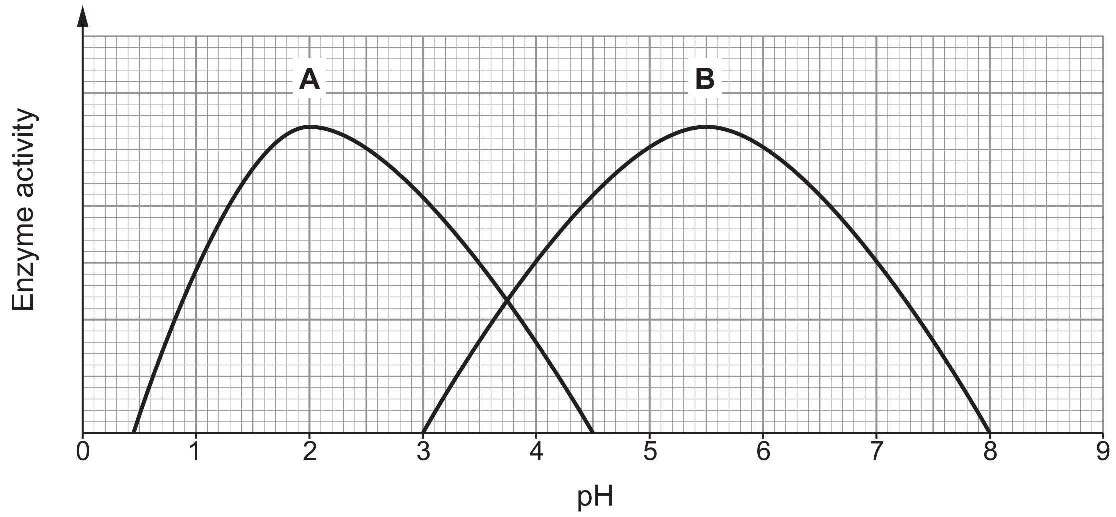
11. *(a)* Enzymes are biological catalysts. State what is meant by the term *catalyst*. [2]

....

....

....

(b) The following graphs show how the activity of two enzymes, **A** and **B**, varies with pH.



(i) Use the graphs to compare the activities of the two enzymes. [2]

....

....

....

(ii) Enzyme **C** is found in saliva. It works between pH 5 and pH 9 but is best at a neutral pH. **Sketch on the grid above** how the activity varies with pH. [2]

6

END OF PAPER

FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	**Formula**	**Name**	**Formula**
aluminium	Al^{3+}	bromide	Br^{-}
ammonium	NH_4^{+}	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^{-}
calcium	Ca^{2+}	fluoride	F^{-}
copper(II)	Cu^{2+}	hydroxide	OH^{-}
hydrogen	H^{+}	iodide	I^{-}
iron(II)	Fe^{2+}	nitrate	NO_3^{-}
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^{+}	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^{+}		
silver	Ag^{+}		
sodium	Na^{+}		
zinc	Zn^{2+}		

THE PERIODIC TABLE

1	2	Group										3	4	5	6	7	0
						1 H Hydrogen 1											4 He Helium 2
7 Li Lithium 3	9 Be Beryllium 4											11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10
23 Na Sodium 11	24 Mg Magnesium 12											27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulfur 16	35.5 Cl Chlorine 17	40 Ar Argon 18
39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36
86 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	99 Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54
133 Cs Caesium 55	137 Ba Barium 56	139 La Lanthanum 57	179 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86
223 Fr Francium 87	226 Ra Radium 88	227 Ac Actinium 89															

Key

A_r	relative atomic mass
Symbol	
Name	
Z	atomic number

Surname	Centre Number	Candidate Number
First name(s)		0

GCSE

3410U10-1

Z22-3410U10-1

FRIDAY, 17 JUNE 2022 – AFTERNOON

CHEMISTRY – Unit 1:
Chemical Substances, Reactions and Essential Resources

FOUNDATION TIER

1 hour 45 minutes

For Examiner's use only		
Question	**Maximum Mark**	**Mark Awarded**
1.	**7**	
2.	**3**	
3.	**6**	
4.	**8**	
5.	**12**	
6.	**8**	
7.	**10**	
8.	**6**	
9.	**5**	
10.	**9**	
11.	**6**	
Total	**80**	

3410U101 01

ADDITIONAL MATERIALS

In addition to this examination paper you will need a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid. You may use a pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

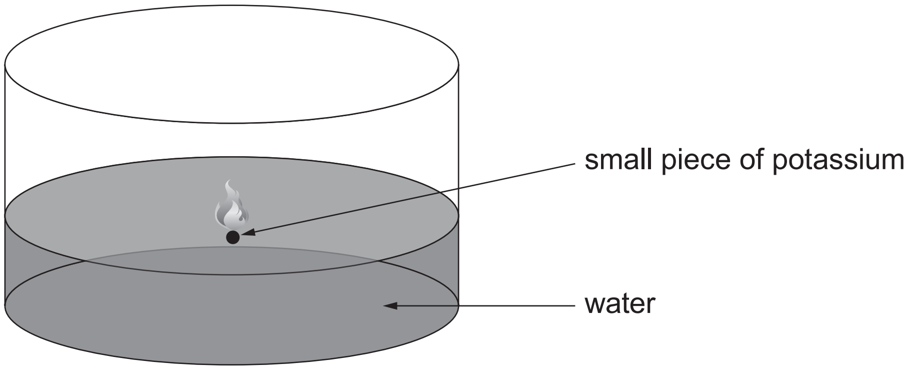
Question **8** is a quality of extended response (QER) question where your writing skills will be assessed.

The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.

Answer **all** questions.

1. (a) Potassium reacts with water to produce potassium hydroxide and hydrogen gas.

$$2K + 2H_2O \longrightarrow 2KOH + H_2$$



Examiner only

(i) Complete the table. Tick (✓) **one** box in **each** of the last two rows. [2]

Substance	Formula	Element	Compound
potassium	K	✓	
water	H_2O		✓
potassium hydroxide	KOH		
hydrogen	H_2		

(ii) During the reaction, potassium burns with a **lilac** flame.

Give **two** other observations that you would expect to make.

Choose your answers from the box. [2]

orange crystals form	**fizzing**	**potassium sinks**
potassium floats	**white precipitate forms**	

Observation 1 ....

Observation 2 ....

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03

Examiner only

(b) Potassium also reacts with oxygen to produce potassium oxide.

$$\boxed{}\,K + O_2 \longrightarrow 2K_2O$$

Choose a number from the box to balance the equation for this reaction. [1]

2	**4**	**6**

(c) Most Group 1 and Group 2 metals can be identified by the colour seen in a flame test.

Draw **one** line from each metal to the colour seen.

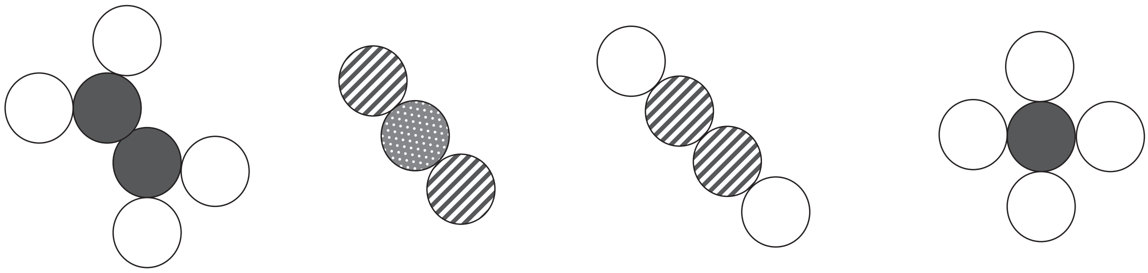
One line has been drawn for you. [2]

Metal	**Colour seen**
	white
lithium	red
sodium	green
barium	blue
	yellow

7

Examiner only

2. (a) The following diagrams represent molecules of some gases.



A B C D

Give the **letter** of the diagram which represents each of the following molecules. [2]

sulfur dioxide, SO_2 ....

ethene, C_2H_4 ....

(b) This chemical equation shows the burning of ethene, C_2H_4.

$$C_2H_4 + 3O_2 \longrightarrow 2CO_2 + 2H_2O$$

Complete the **word** equation for this reaction. [1]

ethene + oxygen $\longrightarrow$ .... + water

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3

Examiner only

3. Chlorine, bromine and iodine are elements in Group 7 of the Periodic Table.

(a) State the number of electrons in the outer shell of a chlorine atom. [1]

....

(b) Underline the correct formula of chlorine gas. [1]

$\mathbf{Cl_2}$ $\mathbf{Cl^2}$ **Cl** **2Cl**

(c) A teacher demonstrated the reactivity of Group 7 elements by placing heated iron wool in gas jars containing vapours of the elements.

The class recorded the results shown in the table.

Group 7 element	Observation with heated iron wool
chlorine	glowed very brightly
bromine	
iodine	glowed faintly

(i) Put a tick (✓) in the box to show the observation when heated iron wool was placed in bromine vapour. [1]

glowed less brightly than iodine ☐

glowed less brightly than chlorine ☐

glowed more brightly than chlorine ☐

Examiner only

(ii) The compound produced in the reaction between iron wool and bromine contains the ions Fe^{3+} and Br^{-}.

I. Give the formula of this compound. [1]

....

II. State the name of this compound. [1]

....

(d) Tick (✓) the box next to a common use of iodine. [1]

to disinfect skin before surgery	☐
to make coloured fireworks	☐
to sterilise swimming pools	☐
to fill party balloons	☐

3410U101 07

6

Examiner only

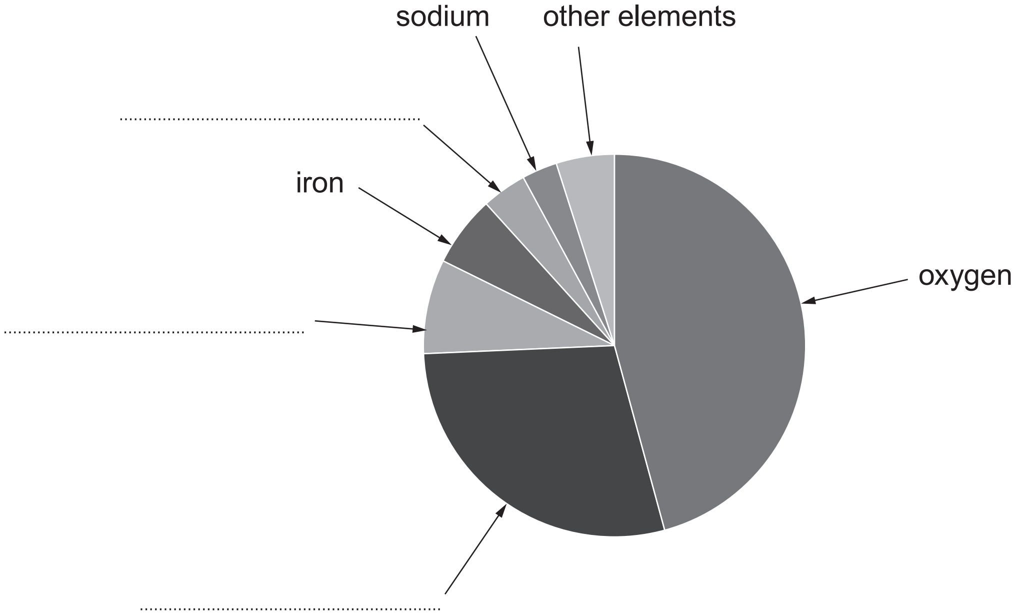
4. (a) The Earth's crust contains more than 80 elements, including many metals. Some of the metals, such as gold, are found 'native' or uncombined, but most are bonded with other elements to form compounds.

The crust can be thicker than 80 kilometres in some spots and less than one kilometre thick in others.

Just six elements make up almost all of the Earth's crust. These are aluminium, calcium, iron, oxygen, silicon and sodium.

Silicon is the second most abundant element. The amount of aluminium is approximately double the amount of calcium.

The chart shows the percentages of elements contained in the crust.



(i) Use the information given to **label** the remaining sections of the chart. [2]

(ii) Underline the approximate fraction of elements in the crust which are metals. [1]

$\frac{1}{2}$ $\frac{1}{3}$ $\frac{1}{4}$ $\frac{3}{4}$

Examiner only

(iii) Tick (✓) the statement which best explains why most metals are **not** found native in the crust. [1]

most metals have higher melting points than gold ☐

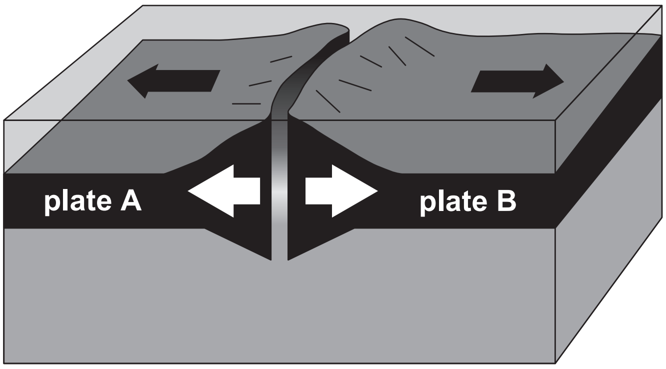
most metals are magnetic ☐

most metals are more reactive than gold ☐

most metals are radioactive ☐

(b) Tectonic plates are huge sections of the crust that move very slowly above the mantle. The places where plates meet are called plate boundaries.

The diagram shows one type of plate boundary.



(i) Describe how new rock is formed at this plate boundary. [3]

....

....

....

....

(ii) Underline the name of this type of plate boundary. [1]

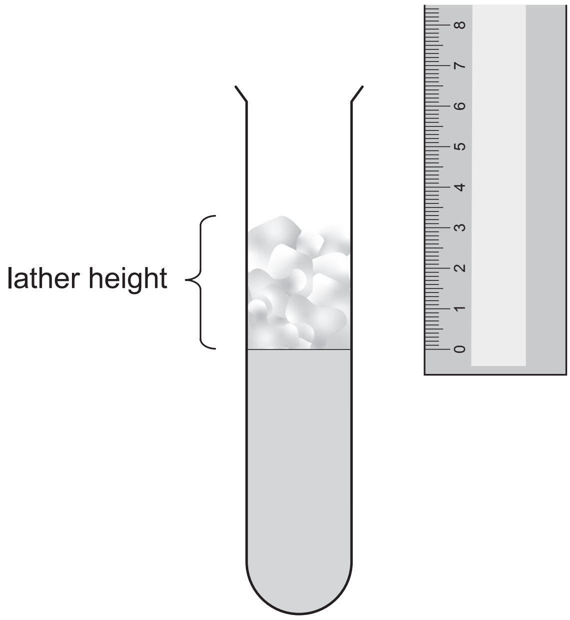
conservative **destructive** **constructive**

8

3410U101 09

5. (a) A class measured the hardness of five different samples of water, **A**, **B**, **C**, **D** and **E**.

They added soap solution to each water sample and measured the height of the lather produced on shaking.



Their results are shown in the table.

Water sample	Lather height (cm)
A	1.5
B	3.0
C	0.5
D	5.0
E	2.0

Examiner only

(i) Complete the order of hardness of the water samples, from softest to hardest. [1]

softest → hardest

D				**C**

(ii) Calcium is one of two metal ions which cause hardness in water.

Underline the name of the other metal ion which causes hardness in water. [1]

potassium **sodium** **nickel** **magnesium** **tin**

(iii) Tick (✓) **all** of the variables which must be controlled in this experiment. [2]

volume of soap solution	☐
type of water	☐
type of soap solution	☐
volume of water	☐
height of lather	☐
width of test tube	☐

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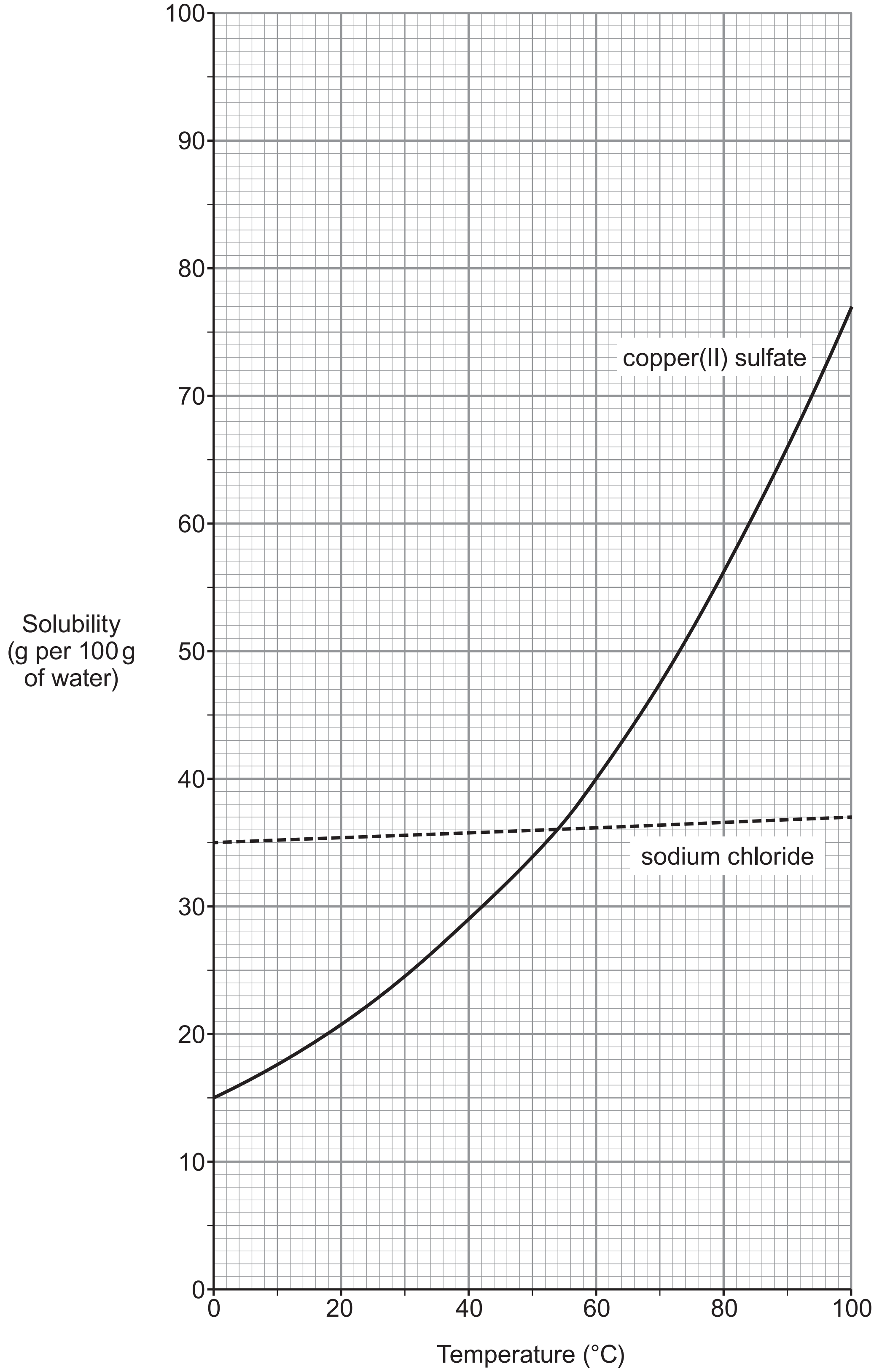
11

(b) The graph shows the solubilities of sodium chloride and copper(II) sulfate at different temperatures.



Examiner only

(i) State the temperature at which the two compounds have the same solubility. [1]

.... °C

(ii) Compare the changes in solubility of the two compounds. [2]

....

....

....

(iii) Calculate the mass of copper(II) sulfate that would dissolve in **1000 g** of water at 40 °C. [2]

Mass = ... g

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13

(c) Potassium sulfate, K_2SO_4, is soluble in water.

(i) State the number of sulfur atoms in the formula K_2SO_4. [1]

....

(ii) Calculate the relative formula mass (M_r) of K_2SO_4. [2]

A_r(K) = 39 A_r(S) = 32 A_r(O) = 16

M_r = ...

12

Examiner only

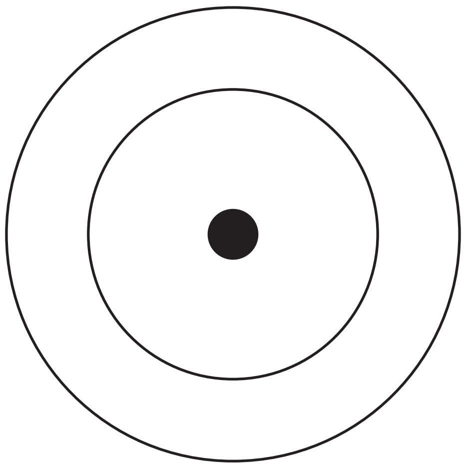
6. (a) An atom of an element has 5 protons, 5 electrons and 6 neutrons.

(i) State the atomic number and the mass number of this atom. [2]

Atomic number ....

Mass number ....

(ii) Complete the diagram of the electronic structure of the element. [1]



(iii) Explain why the atom has no overall charge. [2]

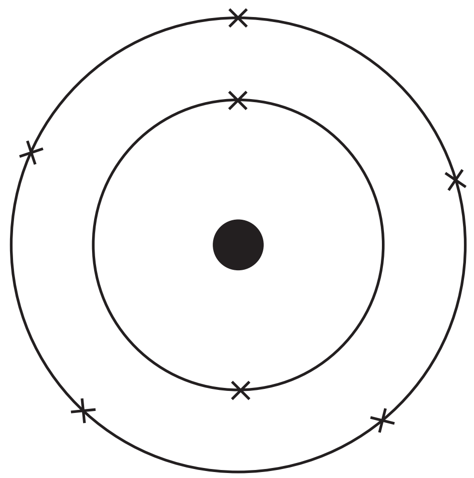
....

....

....

Examiner only

(b) The diagram shows the electronic structure of another element.



This element is in Group 5 and Period 2 of the Periodic Table.

(i) State the name of the element. [1]

....

(ii) Use the electronic structure to explain why this element is in Group 5. [1]

....

....

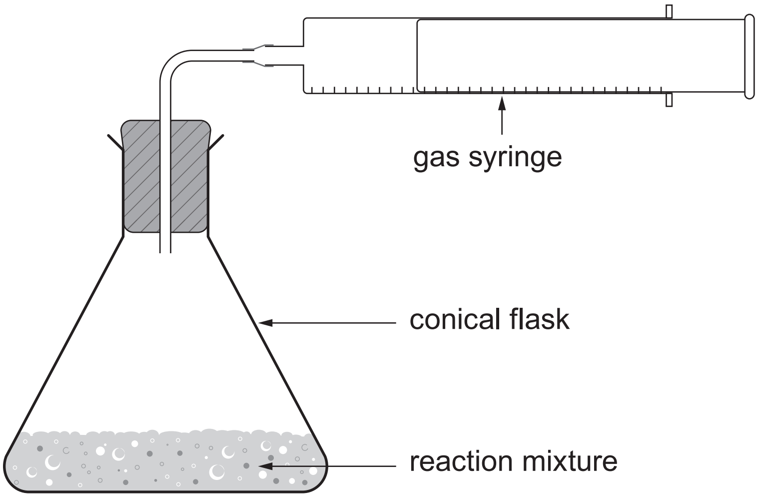
(iii) Use the electronic structure to explain why this element is in Period 2. [1]

....

....

8

7. A student decided to investigate the rate of reaction of different concentrations of hydrochloric acid with pieces of chalk, using the equipment shown.



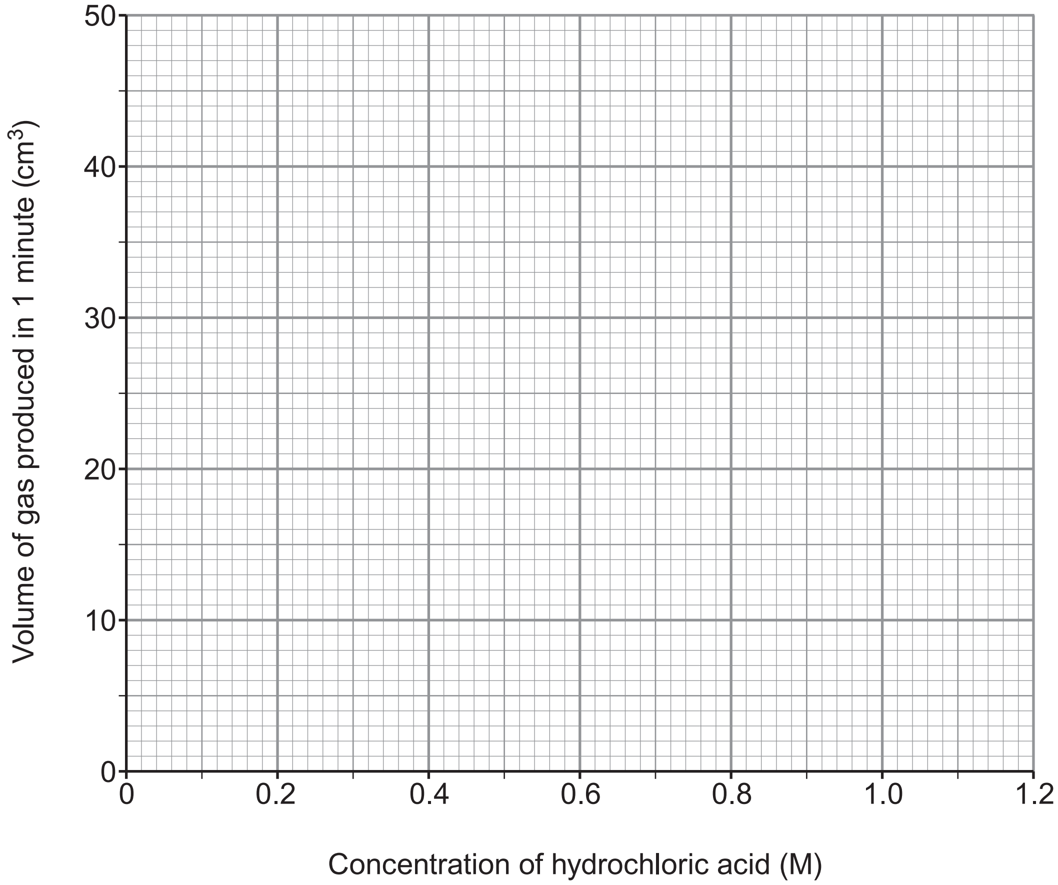
Chalk contains calcium carbonate and the reaction produces carbon dioxide gas.

The table shows the student's results.

Concentration of hydrochloric acid (M)	Volume of gas produced in 1 minute (cm^3)
0.2	8
0.4	16
0.6	24
0.8	32
1.0	40
1.2	48

Examiner only

(a) Plot the results from the table on the grid. Draw a suitable line. [3]



Examiner only

(b) Describe how the volume of gas produced in 1 minute changes as the concentration of the acid changes. [2]

....

....

....

(c) Underline the correct word(s) in each bracket to explain the results. [3]

When the acid concentration is higher, there are

(more / less / the same number of) particles in the same volume.

Acid particles **(dissolve / mix / collide)** with the chalk more frequently.

The **(chalk / acid / gas)** is produced at a higher rate.

(d) Other than changing the concentration of the acid or the mass of the chalk added, state **two** ways in which the rate of this reaction could be increased. [2]

....

....

....

10

Examiner only

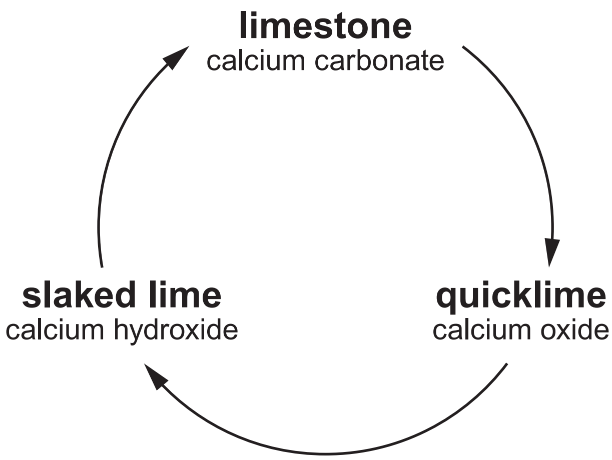
8. Burning fossil fuels leads to global warming.

Explain how global warming arises and describe the problems caused. [6 QER]

6

Examiner only

9. Limestone is a rock which consists mostly of calcium carbonate. The diagram shows a cycle of reactions involving limestone.



(a) (i) When limestone is heated, calcium carbonate is converted to calcium oxide and carbon dioxide.

I. State the name for this type of reaction. [1]

....

II. Write a balanced symbol equation for the reaction. [2]

.... $\longrightarrow$ .... + ...

(ii) State what must be added to calcium oxide to form calcium hydroxide. [1]

....

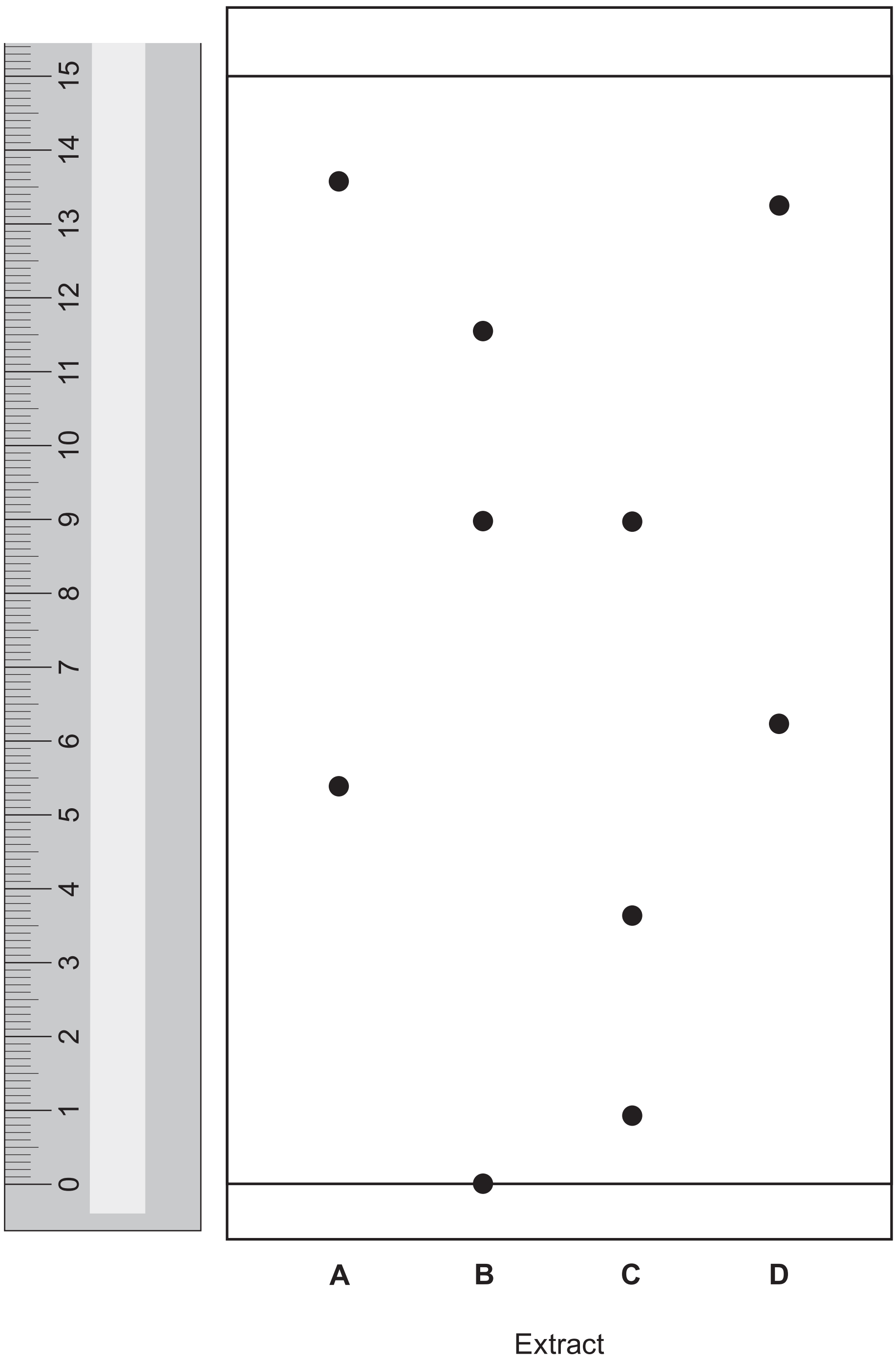
(b) Give **one** use of limestone in the construction industry. [1]

....

5

10. Paper chromatography can be used to identify plant leaf pigments. This process uses a chemical called acetone as the solvent instead of water.

The diagram shows the chromatogram of plant leaf extracts **A**, **B**, **C** and **D** in acetone.



Examiner only

(a) All of the extracts contain a mixture of pigments with different R_f values.

For which plant leaf extract, **A**, **B**, **C** or **D**, is the **highest** R_f value 0.60?

Give your reasoning. [3]

Extract ....

Reasoning ....

....

....

(b) Explain why the pigments travel different distances on the chromatogram. [2]

....

....

....

(c) One of the extracts contains a pigment which is insoluble in acetone.

State the **letter** of this extract. Explain your choice. [2]

Extract ....

Explanation ....

....

(d) The chemical formula of the solvent acetone is C_3H_6O. Calculate the percentage by mass of carbon in acetone.

The relative formula mass (M_r) of acetone is 58. [2]

$A_r(C) = 12$

Percentage = ... %

9

Examiner only

11. (a) The Earth's early atmosphere contained large amounts of water vapour and carbon dioxide. Explain why the amounts of water vapour and carbon dioxide decreased over geological time. [4]

Water vapour ....

....

....

....

Carbon dioxide ....

....

....

....

(b) State the percentages of nitrogen and oxygen in the present atmosphere. [2]

Nitrogen .... %

Oxygen .... %

6

END OF PAPER

FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	**Formula**	**Name**	**Formula**
aluminium	Al^{3+}	bromide	Br^{-}
ammonium	NH_4^{+}	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^{-}
calcium	Ca^{2+}	fluoride	F^{-}
copper(II)	Cu^{2+}	hydroxide	OH^{-}
hydrogen	H^{+}	iodide	I^{-}
iron(II)	Fe^{2+}	nitrate	NO_3^{-}
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^{+}	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^{+}		
silver	Ag^{+}		
sodium	Na^{+}		
zinc	Zn^{2+}		

THE PERIODIC TABLE

Group

1	2											3	4	5	6	7	0
						1 H Hydrogen 1											4 He Helium 2
7 Li Lithium 3	9 Be Beryllium 4											11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10
23 Na Sodium 11	24 Mg Magnesium 12											27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulfur 16	35.5 Cl Chlorine 17	40 Ar Argon 18
39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36
86 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	99 Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54
133 Cs Caesium 55	137 Ba Barium 56	139 La Lanthanum 57	179 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86
223 Fr Francium 87	226 Ra Radium 88	227 Ac Actinium 89															

Key

A_r — relative atomic mass
Symbol
Name
Z — atomic number

Surname	Centre Number	Candidate Number
First name(s)		0

GCSE

3410U10-1

S23-3410U10-1

FRIDAY, 16 JUNE 2023 – MORNING

CHEMISTRY – Unit 1:
Chemical Substances, Reactions and Essential Resources

FOUNDATION TIER

1 hour 45 minutes

For Examiner's use only		
Question	**Maximum Mark**	**Mark Awarded**
1.	**7**	
2.	**8**	
3.	**7**	
4.	**10**	
5.	**9**	
6.	**6**	
7.	**5**	
8.	**8**	
9.	**9**	
10.	**11**	
Total	**80**	

3410U101
01

ADDITIONAL MATERIALS

In addition to this examination paper you will need a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

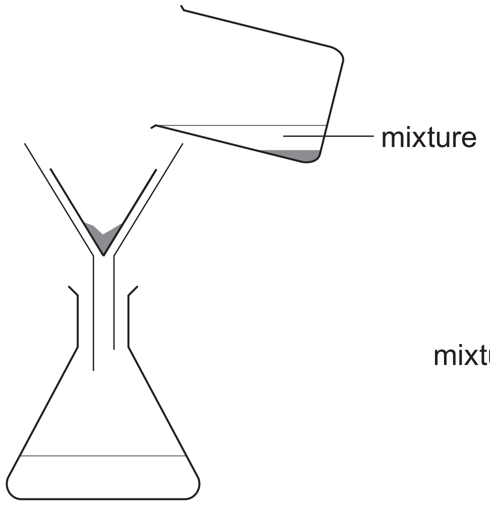
The number of marks is given in brackets at the end of each question or part-question.

The assessment of the quality of extended response (QER) will take place in Question **5**(b).

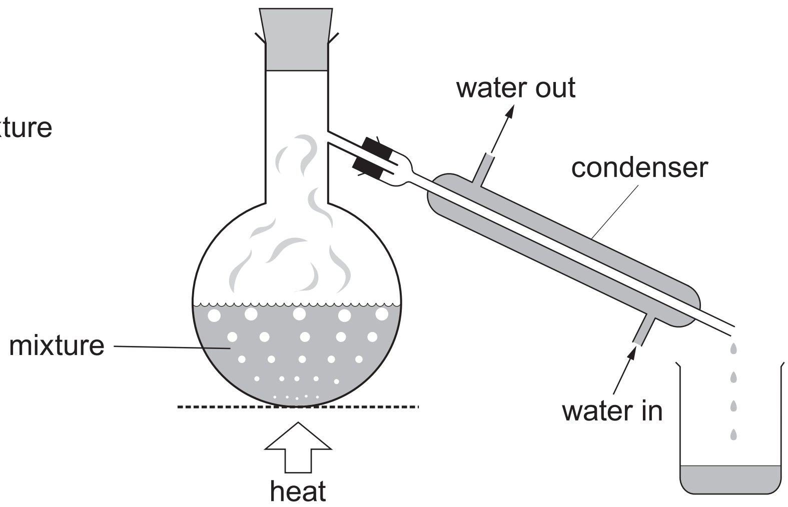
The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.

Answer **all** questions.

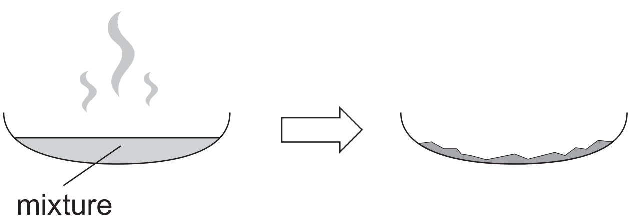
1. (a) The diagrams show four methods, **A**, **B**, **C** and **D**, used to separate different mixtures.



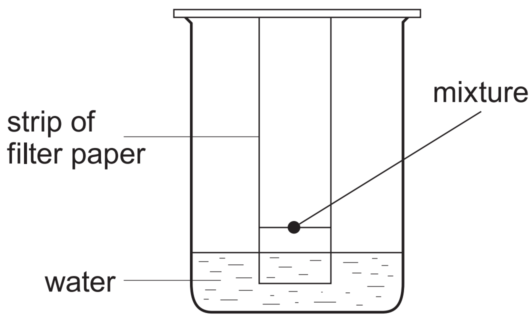
Method **A**



Method **B**



Method **C**



Method **D**

Examiner only

(i) Choose from the box the names of methods **B** and **D**. [2]

distillation	**chromatography**	**filtration**	**evaporation**	**boiling**

Method **B** ....

Method **D** ....

(ii) Give the letter, **A**, **B**, **C** or **D**, of the method used to [3]

remove sand from water ....

obtain pure water from sea water ....

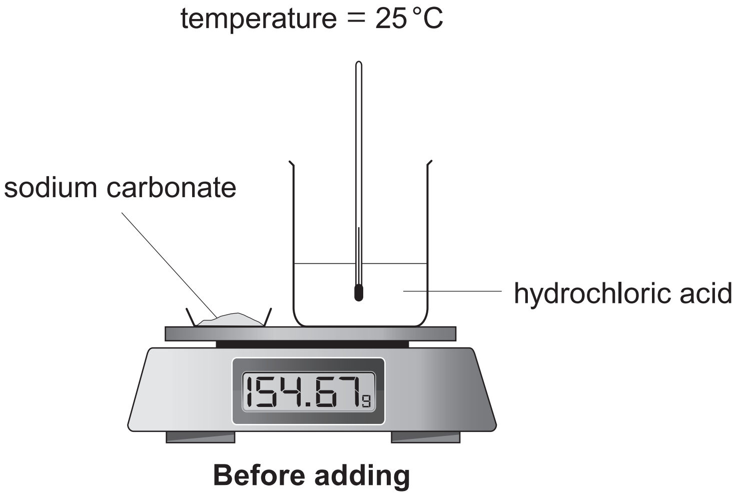
separate red and yellow dyes ....

3410U101 03

Examiner only

(b) Sodium carbonate reacts with dilute hydrochloric acid forming sodium chloride, water and carbon dioxide.

The diagrams show the apparatus before and after sodium carbonate is added to hydrochloric acid.



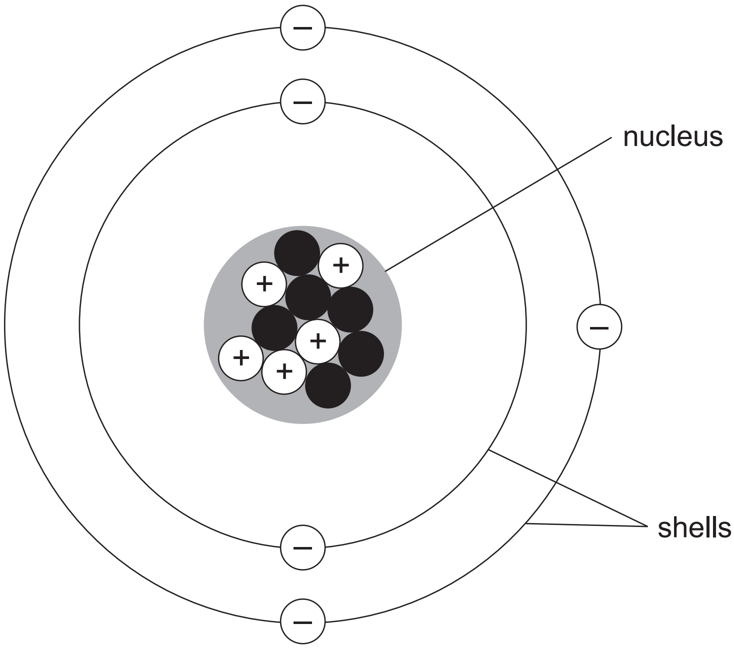
Tick (✓) **two** observations that show a chemical reaction is taking place. [2]

The solid stays the same	☐
A gas is formed	☐
A temperature change occurs	☐
The mass of the beaker and contents stays the same	☐

7

Examiner only

2. (a) Atoms contain particles called protons, neutrons and electrons. The diagram shows a model of an atom of boron.



State whether the statements in the table below are **true** or **false**. [4]

Statement	True or false?
Boron atoms contain the same number of protons and electrons	....
The particles found in the shells are called electrons	....
The nucleus contains five neutrons	....
The electronic structure of boron is 3,2	....

Examiner only

(b) The formula of boron trioxide is B_2O_3.

Calculate the relative formula mass (M_r) of boron trioxide. [2]

$A_r(B) = 11 \qquad A_r(O) = 16$

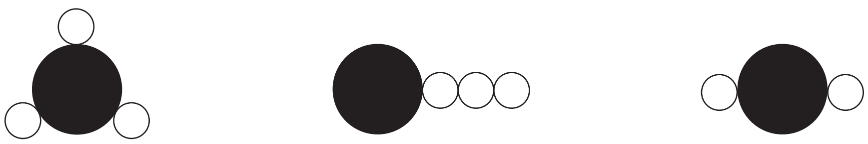
Relative formula mass = ...

(c) The diagrams below represent atoms of boron and fluorine.

boron, B fluorine, F

Boron trifluoride has the formula BF_3.

Choose the **letter** of the diagram that represents a molecule of boron trifluoride. [1]



A **B** **C**

Letter ....

(d) Magnesium fluoride contains the ions Mg^{2+} and F^-.

Underline the correct formula for magnesium fluoride. [1]

Mg2F **Mg_2F** **MgF^2** **MgF_2**

8

3410U101 07

 (3410U10-1) **Turn over.**

Examiner only

3. (a) (i) The diagram below shows the position of the Earth's continents today.

In 1912 Alfred Wegener suggested that all the continents must once have been joined together as one big land mass.

Diagrams **A**, **B** and **C** show the position of the Earth's continents 50 million, 100 million and 150 million years ago, but not necessarily in that order.

A **B** **C**

Give the letter, **A**, **B** or **C**, of the diagram which shows the position of the Earth's continents 150 million years ago. [1]

Letter ....

Examiner only

(ii) Wegener's theory of continental drift was not accepted by other scientists until several years after his death in 1930. The evidence to support his theory was found in 1960 when part of the ocean floor was surveyed around a plate boundary. The table shows data collected from the survey.

Distance of ocean floor from plate boundary (km)	Approximate age of rock (million years)
2000	100

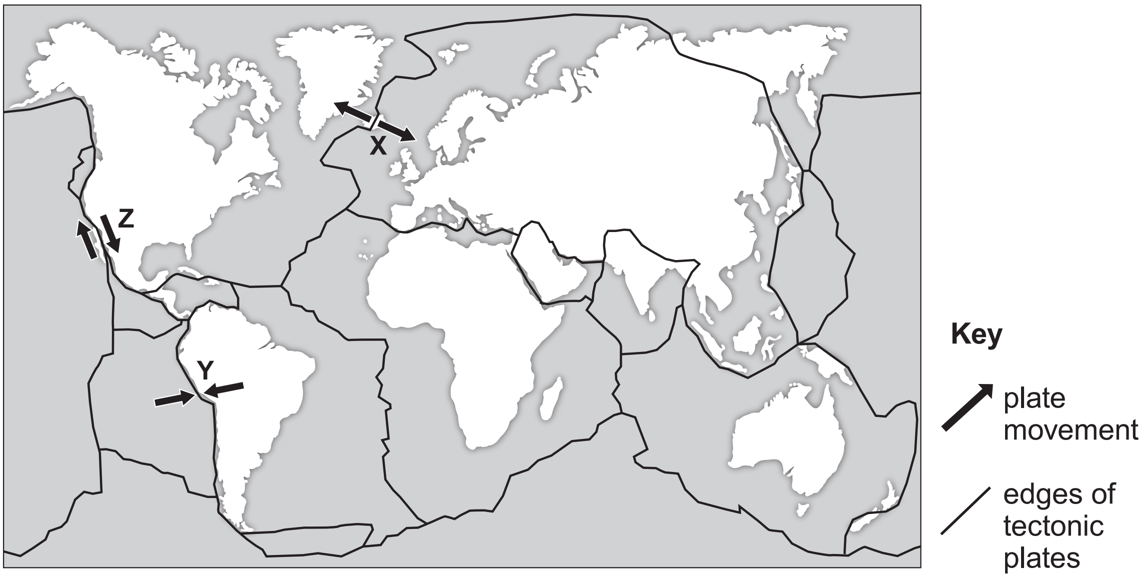
Calculate the mean speed at which the ocean floor is spreading. [1]

$$\text{mean speed (km/million years)} = \frac{\text{distance (km)}}{\text{time (million years)}}$$

Mean speed = ... km/million years

3410U101 09

(iii) The map shows some information about tectonic plates and three locations **X**, **Y** and **Z**.



Give the **letter** of the location you would expect to have earthquakes but not volcanic eruptions. [1]

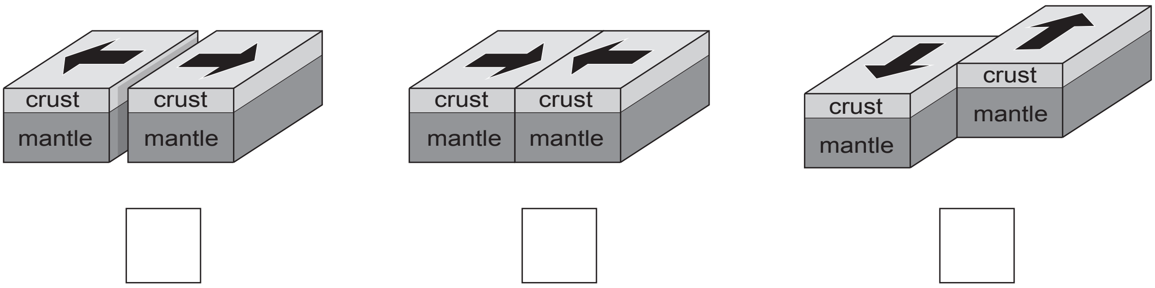
Letter ....

Examiner only

(b) The photograph below shows ‘pillow lava’ which was formed from volcanoes on the sea bed at a **constructive** plate boundary millions of years ago.

‘pillow lava’ on Llanddwyn Island, Anglesey

(i) Tick (✓) the box of the diagram that shows a constructive plate boundary where the pillow lava was formed. [1]



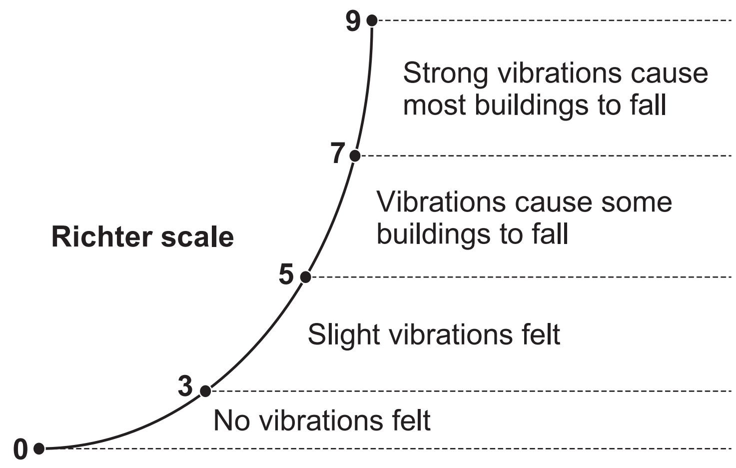
(ii) Complete the sentences by underlining the correct word(s) in the brackets. [2]

Pillow lava is formed at a constructive plate boundary when **(magma / sea water / crust)** rises and cools, forming new rock.

The movement of the Earth’s tectonic plates is caused by **(electric currents / convection currents / ocean currents)** within the mantle.

Examiner only

(c) Charles Richter developed the Richter Scale in 1935 to measure the strength of earthquakes.



In June 2018 an earthquake occurred in the Caernarfon area, with a minor tremor being felt.

Circle the number that best shows the size of the earthquake in Caernarfon. [1]

1 **4** **6** **8**

3410U101
11

7

Examiner only

4. Dilute hydrochloric acid reacts with magnesium forming magnesium chloride and hydrogen gas.

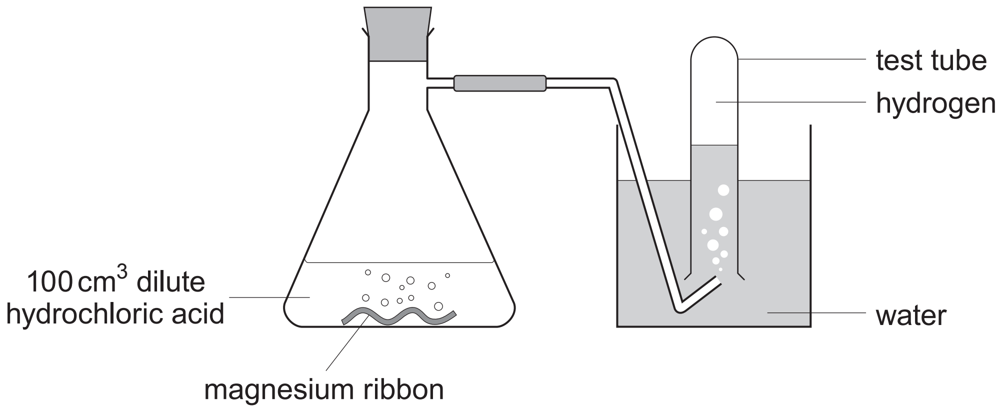
(a) Tick (✓) the box next to the correct equation for the reaction between magnesium and hydrochloric acid. [1]

$Mg + HCl \longrightarrow MgCl_2 + H$ ☐

$Mg + 2HCl \longrightarrow MgCl_2 + 2H$ ☐

$Mg + 2HCl \longrightarrow MgCl_2 + H_2$ ☐

(b) Osian wanted to find out how changing the concentration of the acid affects the rate of the reaction. He carried out five experiments at room temperature (20 °C). He added a 4 cm piece of magnesium ribbon to 100 cm³ of hydrochloric acid of five different concentrations. He recorded the time it took to half-fill a test tube with gas.



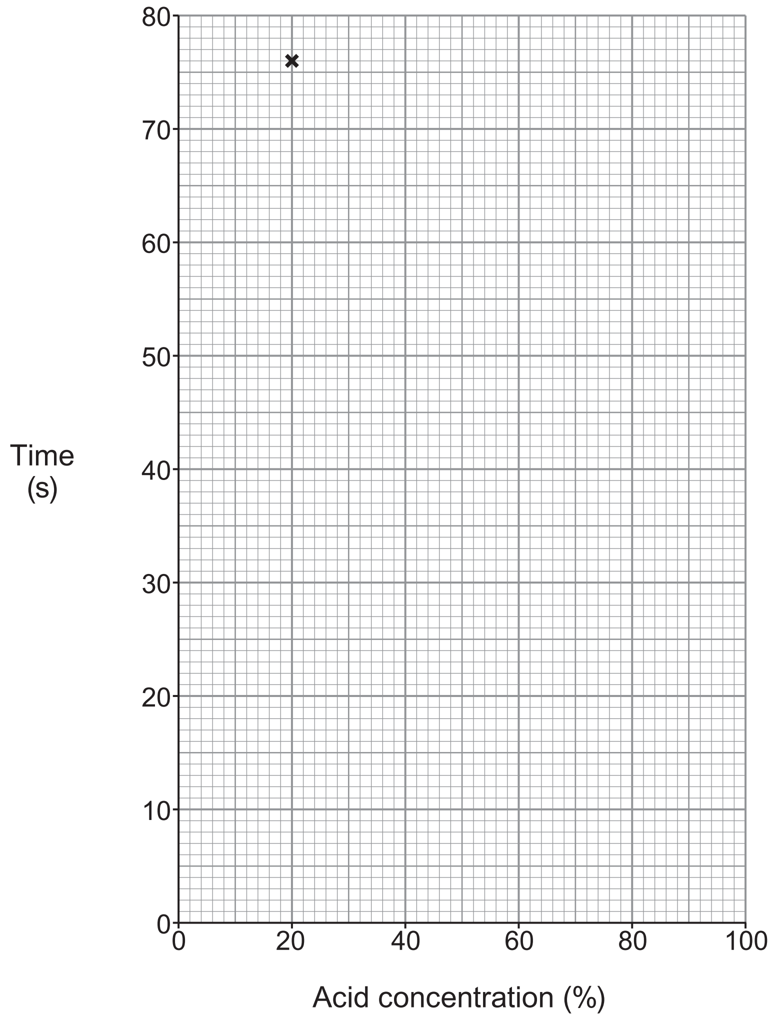
His results are shown below.

Experiment	Acid concentration (%)	Time (s)
1	100	12
2	80	14
3	60	20
4	40	36
5	20	76

Examiner only

(i) Plot the acid concentration against time on the grid below and draw a suitable line.

One point has been plotted for you. [3]



(ii) Underline the correct word(s) in the brackets to complete the following sentences. [2]

As the acid concentration increases, the **time** to half-fill the test tube with gas **(increases / stays the same / decreases)**.

As the acid concentration increases, the **rate** of the reaction **(increases / stays the same / decreases)**.

Examiner only

(iii) Using your knowledge of particle theory, underline the correct words in the brackets to complete the following sentence. [2]

At a higher concentration, there are **(more / less / the same number of)** particles present so there will be **(an equal / a smaller / a greater)** chance of collision.

(iv) There are other ways the rate of the reaction can be changed.

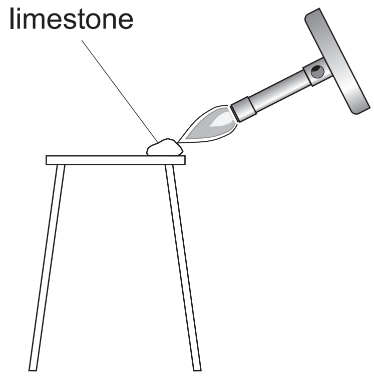
Tick (✓) the **two** statements that correctly describe other ways the rate of reaction can be increased. [2]

Increasing the temperature of the acid	☐
Using a lump of magnesium	☐
Using a different apparatus	☐
Using magnesium powder	☐
Decreasing the temperature of the acid	☐

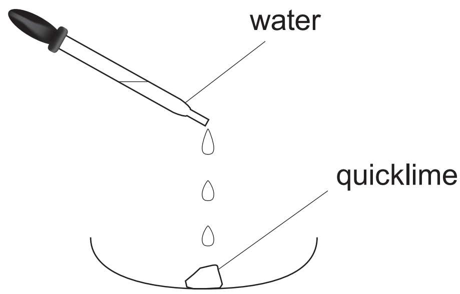
10

Examiner only

5. (a) A student carried out a two-stage experiment to change limestone (calcium carbonate) into slaked lime (calcium hydroxide).



Stage **1**: Limestone (calcium carbonate) decomposes into quicklime (calcium oxide) and carbon dioxide



Stage **2**: Quicklime (calcium oxide) reacts with water forming slaked lime (calcium hydroxide)

(i) Write the formulae for calcium oxide and carbon dioxide to complete the equation for the reaction taking place in stage **1**. [2]

$CaCO_3 \longrightarrow$ .... + ...

(ii) Calcium hydroxide contains one Ca^{2+} ion for every two OH^{-} ions.

Write the chemical formula for calcium hydroxide. [1]

....

 (3410U10-1)

Examiner only

(b)

Describe the economic benefits and environmental drawbacks of limestone quarrying. [6 QER]

....

....

....

....

....

....

....

....

....

....

....

....

....

....

....

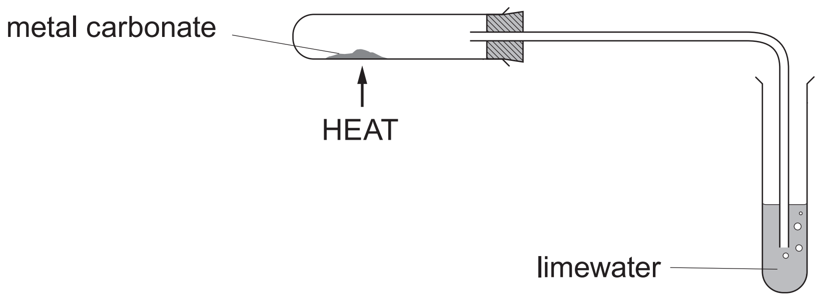
3410U101
17

9

Examiner only

6. (a) Rhian investigated the decomposition of three different metal carbonates.

She measured the time taken for limewater to turn milky using the following apparatus.



Her results are shown in the table.

Metal carbonate	Time taken for limewater to turn milky (s)
copper(II) carbonate	18
zinc carbonate	27
lead carbonate	11

(i) Place the carbonates in order of stability. [1]

Most stable ....

....

Least stable ....

Examiner only

(ii) If sodium carbonate was used in the investigation the limewater would not turn milky however long it was heated.

Tick (✓) the reason why the limewater would not turn milky. [1]

Sodium carbonate only decomposes a small amount on heating	☐
Sodium carbonate is very unstable	☐
Sodium carbonate does not decompose on heating	☐
Sodium carbonate decomposes too quickly	☐

(iii) On heating copper(II) carbonate, Rhian expected to make 5.0 g of copper(II) oxide. She actually made 3.5 g.

Use the formula below to calculate the percentage yield of copper(II) oxide in her experiment. [2]

$$\text{percentage yield} = \frac{\text{actual mass}}{\text{expected mass}} \times 100$$

Percentage yield = ... %

(iv) One of the ions present in copper(II) carbonate is CO_3^{2-}. [1]

Give the formula of the other ion present.

....

(b) Rhian carried out a flame test to show that sodium carbonate contains sodium ions.

Give the colour of the flame seen. [1]

....

6

7. Is it right to waste helium on party balloons?

Helium is a colourless inert gas found in Group 0 of the Periodic Table.

Helium is one of the commonest elements in the Universe, second only to hydrogen. However, on Earth it is relatively rare, as shown in **Table 1**.

Gases which have a density less than air can escape the Earth's gravity and leak away into space. The density of air is 1.2 g/m^3. **Bar chart 1** shows the densities of Group 0 gases.

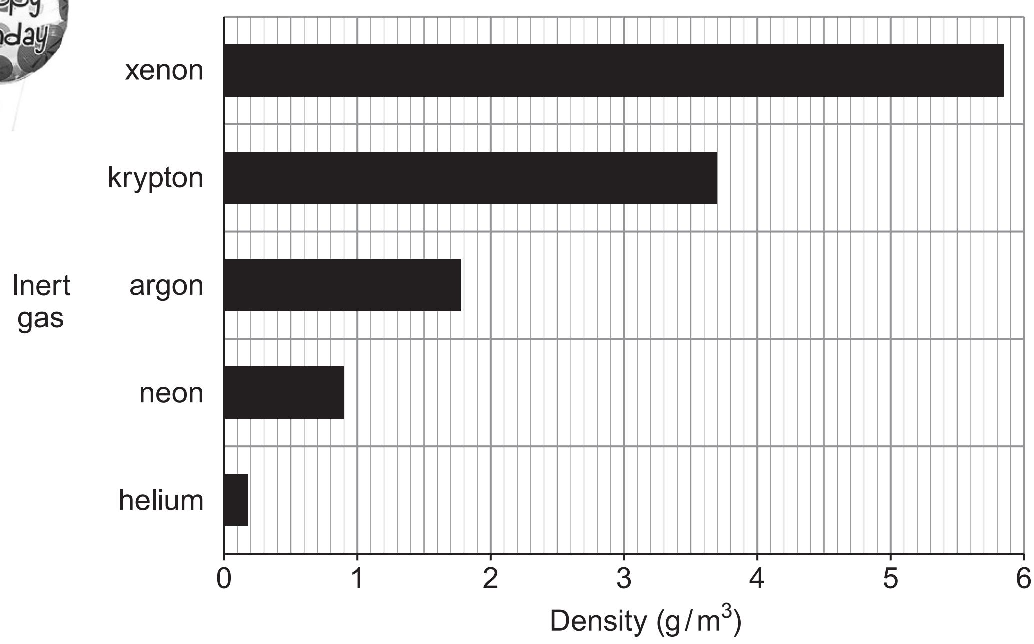
Helium has the lowest boiling point of any element. This makes it of key importance for magnets used in hospital MRI scanners, which must be super-cooled to generate the hugely powerful magnetic fields required.

Some scientists believe that because helium is a finite resource it should not be used for party balloons.

Table 1

Inert gas	Percentage in the atmosphere (%)	Melting point (°C)	Boiling point (°C)
helium	0.00052	−272	−269
neon	0.0018	−246	−246
argon	0.93	−186	−186
krypton	0.0001	−152	−152
xenon	0.000009	−111	−106

Bar chart 1



Examiner only

(a) Answer the following questions using the information given.

(i) Tick (✓) the box next to the **most** important property that makes helium a suitable material to fill **floating** party balloons. [1]

Helium is a gas ☐

Helium is the second most common element in the Universe ☐

Helium is less dense than air ☐

Helium is colourless ☐

(ii) Tick (✓) the box next to the correct statement. [1]

The Earth's atmosphere contains more helium than argon ☐

The Earth's atmosphere contains more xenon than helium ☐

The Earth's atmosphere contains more helium than krypton ☐

(iii) Tick (✓) the box next to the **best** reason for not using helium to fill party balloons. [1]

There isn't much helium in the Earth's atmosphere ☐

Scientists say helium shouldn't be used to fill balloons ☐

Helium is a finite resource ☐

(iv) Tick (✓) the box next to the correct statement. [1]

Only helium gas can leak away into space ☐

Helium and neon gases can leak away into space ☐

Only argon can leak away into space ☐

All inert gases can leak away into space ☐

Examiner only

(b) The table below shows the electronic structure of three Group 0 elements.

Group 0 element	Electronic structure
helium	2
neon	2,8
argon	2,8,8

Tick (✓) the box next to the statement that **best** explains why Group 0 elements are unreactive. [1]

All Group 0 elements have 2 electrons in their inner shell ☐

All Group 0 elements have 8 electrons in their outer shell ☐

All Group 0 elements have full outer shells ☐

All Group 0 elements have some full shells ☐

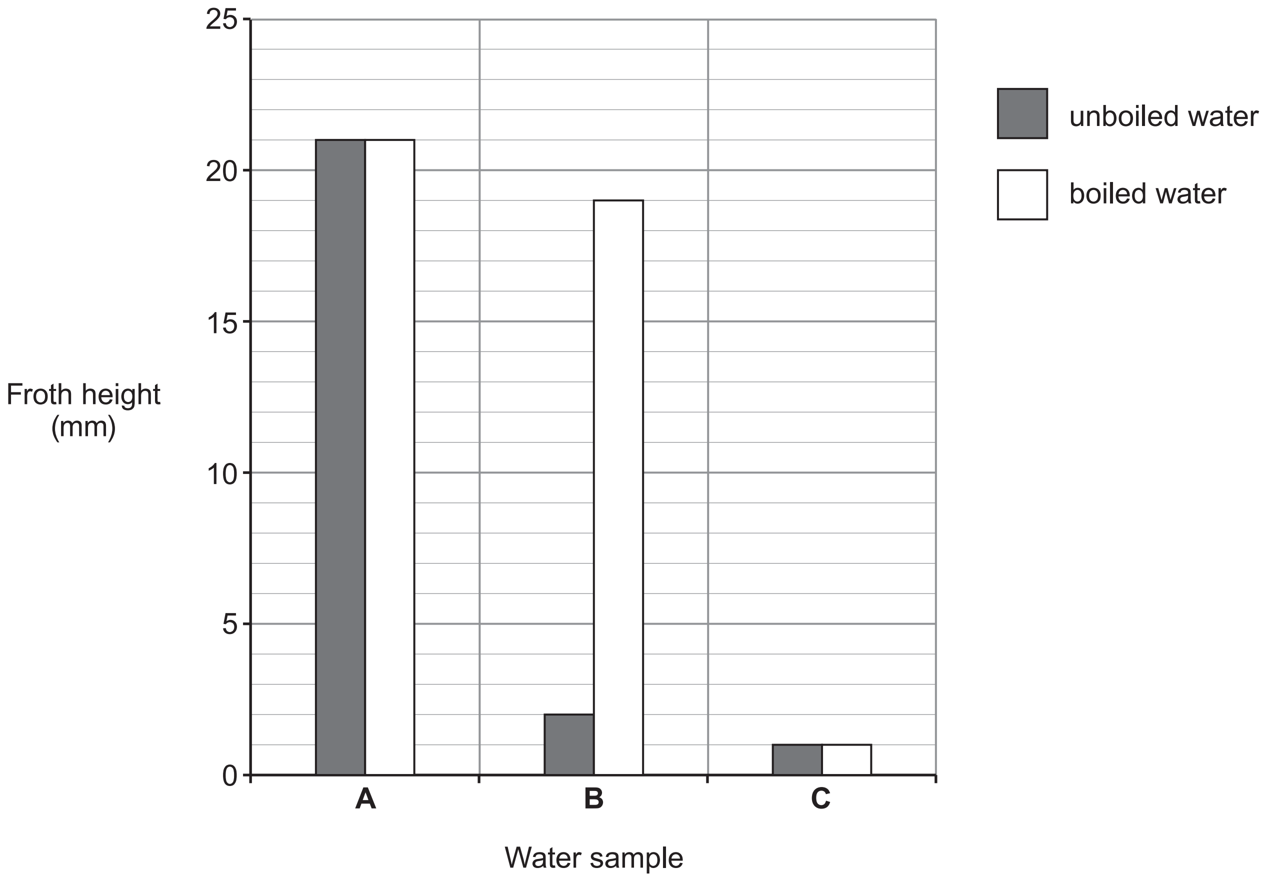
5

Examiner only

8. (a) Three samples of water, **A**, **B** and **C**, from different parts of the UK were tested in a laboratory.

1 cm^3 of soap solution was added to 25 cm^3 of the three different water samples. Each sample was shaken for 1 minute. The height of the froth was measured.

The experiment was repeated using new samples of water, **A**, **B** and **C**, that had been boiled.



Give the **letter** of the water sample which is [2]

temporary hard water ....

permanent hard water ....

soft water ....

Examiner only

(b) There are advantages and disadvantages of living in a hard water area.

Give **two** disadvantages of living in a hard water area. [2]

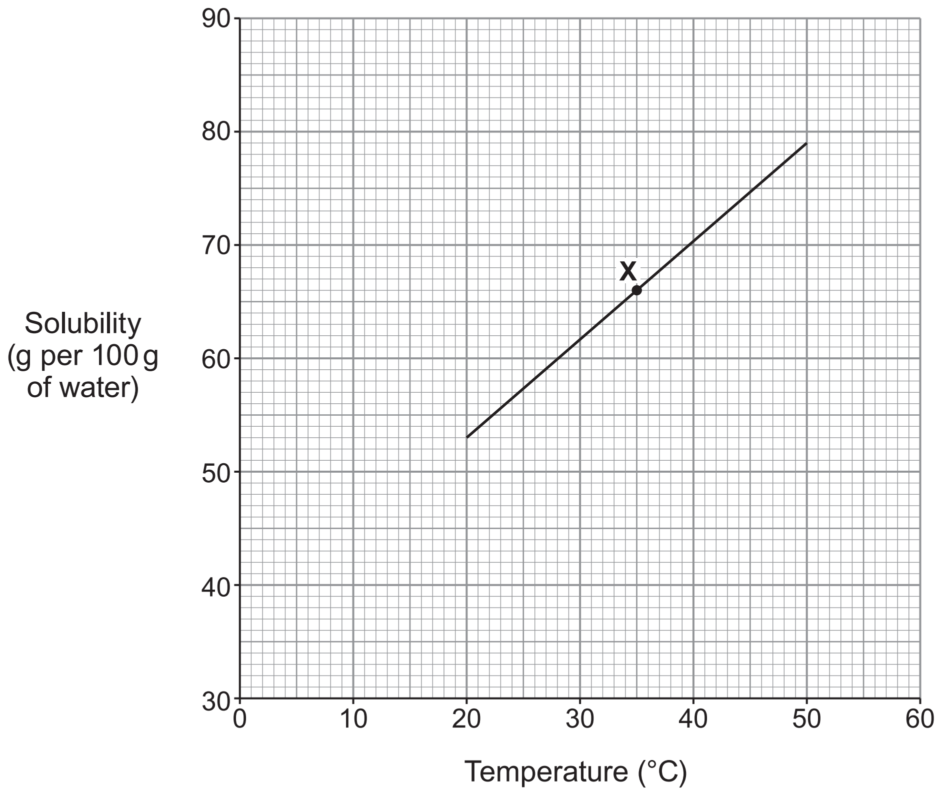
1 ....

....

2 ....

....

(c) The graph below shows the solubility of lead nitrate in water at different temperatures.



(i) State what point **X** on the graph tells you about lead nitrate. [1]

....

....

 (3410U10-1)

Examiner only

(ii) The solubility of lead nitrate at 20 °C is 53 g per 100 g of water.

Use the graph to find its solubility at 50 °C and hence calculate the mass of lead nitrate crystals that form when a saturated solution containing 100 g of water cools from 50 °C to 20 °C. [2]

Mass = ... g

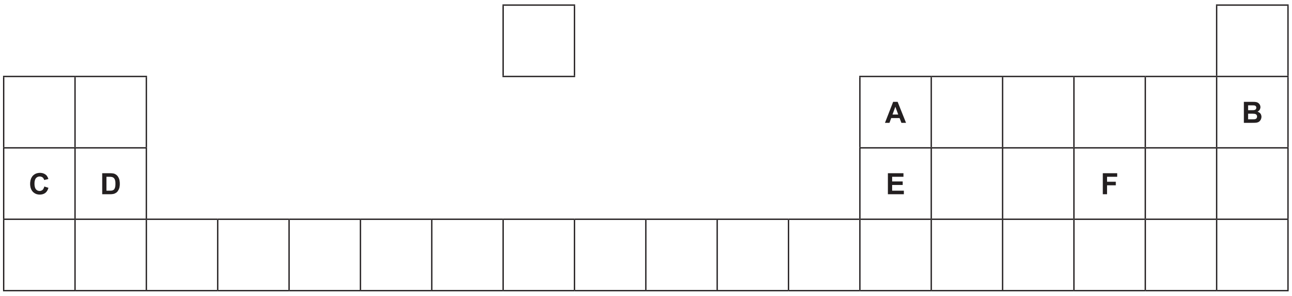
(iii) Use the graph to find the solubility of lead nitrate at 5 °C. [1]

.... g per 100 g of water

8

9. (a) The following diagram shows an outline of part of the Periodic Table.

The letters shown are NOT the chemical symbols of the elements.

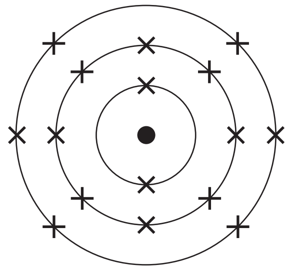


Choose **letters** from the diagram to complete the table below. [4]

	Letter
The element in Group 3 and Period 2	....
The element which has 10 protons in its nucleus	....
The element with the electronic structure 2,8,6	....
The element which forms a 2+ ion	....

Examiner only

(b) The diagram below shows the electronic structure of an element in the Periodic Table.



In the space below, draw a diagram to show the electronic structure of the element which lies directly **above** it. [1]

(c) The table shows information about atoms **X**, **Y** and **Z**.

Atom	Symbol	Number of protons	Number of neutrons	Number of electrons
X	$^{31}_{15}\mathbf{X}$	....	16	15
Y	$^{39}_{19}\mathbf{Y}$	19	....	19
Z	$^{40}_{19}\mathbf{Z}$	19	21	....

(i) Complete the table. [3]

(ii) Underline the term used to describe atoms **Y** and **Z**. [1]

ions **inert** **insoluble** **isotopes**

9

Examiner only

10. (a) The table shows information about some Group 1 elements.

Element	Relative atomic mass	Number of electrons in the outer shell	Melting point (°C)	Boiling point (°C)	Density (g/cm^3)
lithium	7	1	180	1342	0.53
sodium	23	1	98	883	0.97
potassium	39	1	63	759	0.89
rubidium	85	1	39	688	1.53
caesium	134	1	29	671	1.93

Use the information in the table to answer parts (i) and (ii).

(i) State the information which explains why the elements have similar chemical properties. [1]

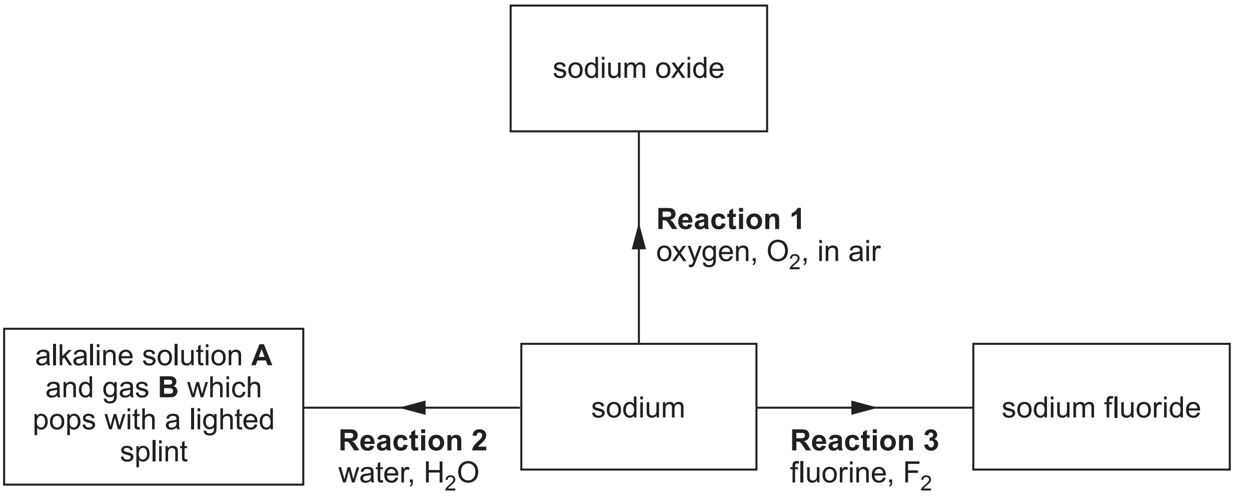
....

(ii) State which **property** has a value which does **not** fit the trend down the group. [1]

....

Examiner only

(b) The flow diagram shows some reactions of sodium.



(i) State how **Reaction 1** is prevented when storing sodium in the laboratory. [1]

....

(ii) Give the names of alkaline solution **A** and gas **B**. [2]

.... and ....

(iii) Name the Group 1 metal which would react **least** violently with water. [1]

....

(iv) Complete the symbol equation for **Reaction 3**. [1]

$$2Na + F_2 \longrightarrow 2 \ldots\ldots\ldots\ldots$$

Examiner only

(c) Sodium fluoride is added to some UK public water supplies to reduce tooth decay in children.

In America sodium hexafluorosilicate, Na_2SiF_6, is more commonly used. The relative formula mass of sodium hexafluorosilicate is 188.

(i) Calculate the percentage of fluorine in sodium hexafluorosilicate. [2]

$A_r(F) = 19 \qquad M_r(Na_2SiF_6) = 188$

Percentage = ... %

(ii) State an **ethical** reason why some people oppose the fluoridation of water supplies. [1]

....

....

(iii) Apart from water supplies, state the most commonly used source of fluoride to reduce tooth decay. [1]

....

11

END OF PAPER

FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	**Formula**	**Name**	**Formula**
aluminium	Al^{3+}	bromide	Br^{-}
ammonium	NH_4^{+}	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^{-}
calcium	Ca^{2+}	fluoride	F^{-}
copper(II)	Cu^{2+}	hydroxide	OH^{-}
hydrogen	H^{+}	iodide	I^{-}
iron(II)	Fe^{2+}	nitrate	NO_3^{-}
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^{+}	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^{+}		
silver	Ag^{+}		
sodium	Na^{+}		
zinc	Zn^{2+}		

THE PERIODIC TABLE

1	2	Group										3	4	5	6	7	0
						1 H Hydrogen 1											4 He Helium 2
7 Li Lithium 3	9 Be Beryllium 4											11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10
23 Na Sodium 11	24 Mg Magnesium 12											27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulfur 16	35.5 Cl Chlorine 17	40 Ar Argon 18
39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36
86 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	99 Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54
133 Cs Caesium 55	137 Ba Barium 56	139 La Lanthanum 57	179 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86
223 Fr Francium 87	226 Ra Radium 88	227 Ac Actinium 89															

Key

A_r — relative atomic mass
Symbol
Name
Z — atomic number